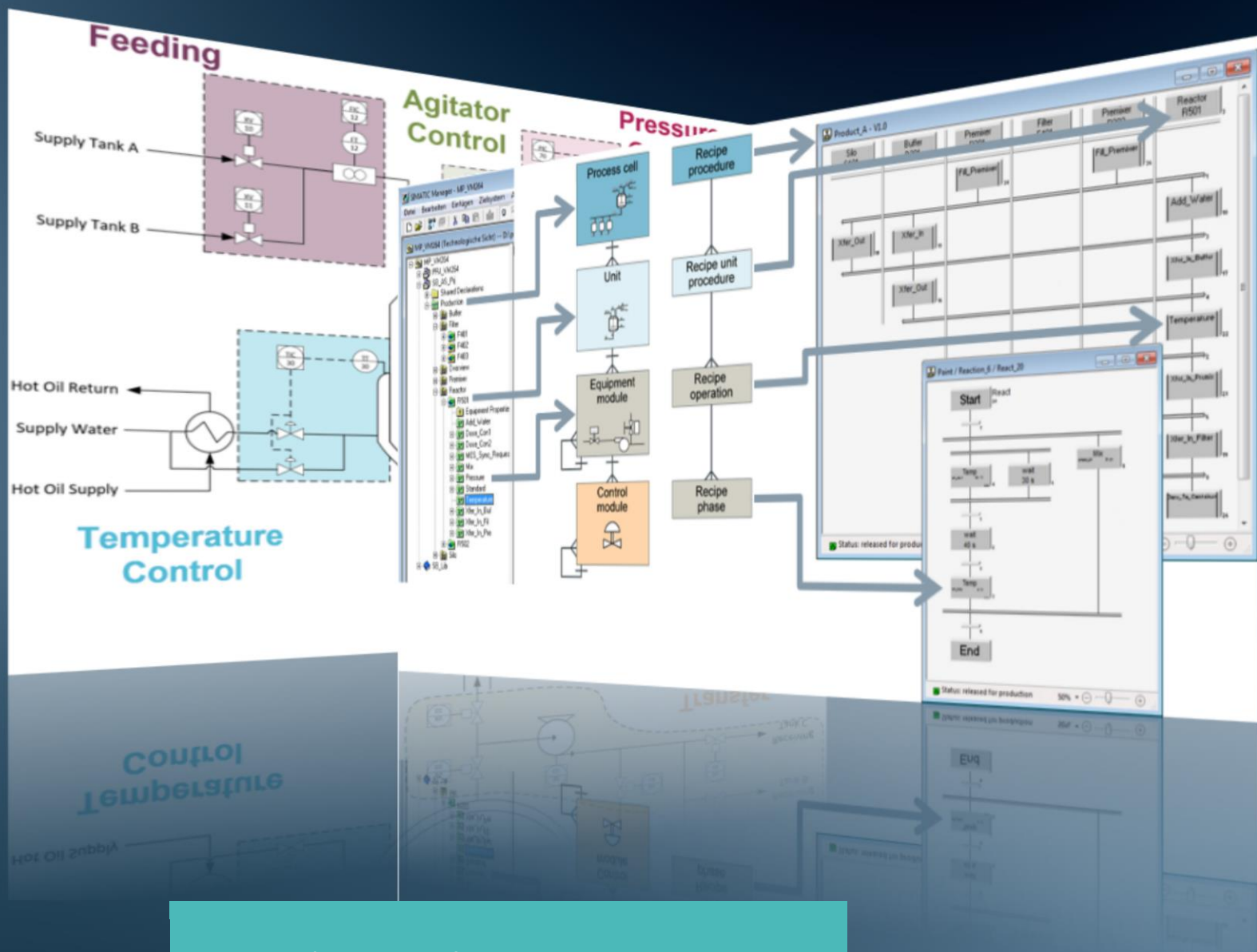




Engineering and automation of batch processes with PCS 7 along ISA-88 models

SIMATIC PCS 7 / SIMATIC BATCH

<https://support.industry.siemens.com/cs/ww/en/view/109784331>

Siemens
Industry
Online
Support



Legal information

Use of application examples

Application examples illustrate the solution of automation tasks through an interaction of several components in the form of text, graphics and/or software modules. The application examples are a free service by Siemens AG and/or a subsidiary of Siemens AG ("Siemens"). They are non-binding and make no claim to completeness or functionality regarding configuration and equipment. The application examples merely offer help with typical tasks; they do not constitute customer-specific solutions. You yourself are responsible for the proper and safe operation of the products in accordance with applicable regulations and must also check the function of the respective application example and customize it for your system.

Siemens grants you the non-exclusive, non-sublicensable and non-transferable right to have the application examples used by technically trained personnel. Any change to the application examples is your responsibility. Sharing the application examples with third parties or copying the application examples or excerpts thereof is permitted only in combination with your own products. The application examples are not required to undergo the customary tests and quality inspections of a chargeable product; they may have functional and performance defects as well as errors. It is your responsibility to use them in such a manner that any malfunctions that may occur do not result in property damage or injury to persons.

Disclaimer of liability

Siemens shall not assume any liability, for any legal reason whatsoever, including, without limitation, liability for the usability, availability, completeness and freedom from defects of the application examples as well as for related information, configuration and performance data and any damage caused thereby. This shall not apply in cases of mandatory liability, for example under the German Product Liability Act, or in cases of intent, gross negligence, or culpable loss of life, bodily injury or damage to health, non-compliance with a guarantee, fraudulent non-disclosure of a defect, or culpable breach of material contractual obligations. Claims for damages arising from a breach of material contractual obligations shall however be limited to the foreseeable damage typical of the type of agreement, unless liability arises from intent or gross negligence or is based on loss of life, bodily injury or damage to health. The foregoing provisions do not imply any change in the burden of proof to your detriment. You shall indemnify Siemens against existing or future claims of third parties in this connection except where Siemens is mandatorily liable.

By using the application examples, you acknowledge that Siemens cannot be held liable for any damage beyond the liability provisions described.

Other information

Siemens reserves the right to make changes to the application examples at any time without notice. In case of discrepancies between the suggestions in the application examples and other Siemens publications such as catalogs, the content of the other documentation shall have precedence.

The Siemens terms of use (<https://support.industry.siemens.com>) shall also apply.

Security information

Siemens provides products and solutions with Industrial Security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the Internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial security measures that may be implemented, please visit <https://www.siemens.com/industrialsecurity>.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed at: <https://www.siemens.com/industrialsecurity>.

Table of contents

Legal information	2
1 Scope of this document	4
2 ISA-88 introduction.....	5
3 Models of the ISA-88	7
3.1 Physical Model	7
3.2 Procedural Control Model.....	9
3.3 Process Model.....	11
4 From Process Flow Diagram to P&ID	14
5 Process Cell Classification.....	16
6 Unit classification	17
7 Control Module and Equipment Module classification.....	18
8 Generic Equipment Module versus recipe-aware Equipment Module approach.....	21
9 The implementation of Equipment Modules in PCS 7	24
9.1 PCS 7 interpretation of EMs and EPHs	24
9.2 What are EMTs and EPHTs?	26
9.3 Design guidelines (comparison of both concepts)	27
9.4 Principal approach.....	28
9.5 Derived EPH and EPH interface	29
9.6 Shared EM	29
10 The implementation of Control Modules in PCS 7	30
10.1 PCS 7 interpretation of Control Modules.....	30
10.2 Benefit	32
10.3 Principal approach.....	32
10.4 Shared Control Module	33
11 Implementation of the Procedural Control Model with Recipes	35
11.1 Batch equipment engineering	35
11.2 Recipe engineering	36
11.3 Separation of procedure and formula.....	38
11.4 Coordination of Units.....	38
11.5 Late binding strategies	39
11.6 Monitoring area	41
11.7 Command step	41
11.8 Batch operation	42
11.9 Batch Online Structural Changes	43
12 Recipe transformation and transferability	44
13 Interfaces towards other systems	46
14 Acronyms	48
15 Appendix	49
15.1 Service and support	49
15.2 Industry Mall	50
15.3 References, links and literature.....	50
15.4 Change documentation	51

1 Scope of this document

Whether producing polymers, beverages or pharmaceuticals – batch processes are indispensable in many industry sectors. In terms of process control and automation, batch production must meet completely different requirements than a continuous process. Selecting a suitable automation philosophy is crucial for the efficiency, flexibility and profitability of the process.

The batch market comprises a wide variety of process types and plant sizes – there is no “one-size-fits-all” solution. This document however should provide an overview along the ISA-88 models of how a project can be modeled and engineered with the capabilities provided by PCS 7. Customers, system integrators or Siemens project personnel shall get an impression what kind of tools are available and how they can be deployed in the plants process control environment.

For more detailed information and guidance on how the different PCS 7 components are utilized in the project implementation the reader should refer to the PCS 7 manuals and compendiums.

2 ISA-88 introduction

Industrial processes can be classified as continuous, discrete or batch manufacturing processes. Continuous processes are classified as processes which have a continuous outflow, for example production of paper or steel. Discrete processes have a discrete output, for example the production of a car.

A batch process is a process that leads to the production of finite quantities of material by subjecting quantities of input materials to an ordered set of processing activities over a finite period of time using one or more pieces of equipment.

There was a need to provide a standard for managing batch processes because of the following reasons:

- No universal model for batch control
- Difficult for plant operators to communicate processing requirements
- Integration of solutions from different vendors is difficult
- Batch control solutions are difficult to configure

Therefore, the ISA (International Society of Automation) decided to provide a standard which describes a method to structure batch processes. It defines terminology and models that makes design and operation of batch plants easier and uniform. This standard is named ISA-88, whereby only the first part of the standard (ANSI/ISA-88.00.01-2010), called "Models and Terminology" is considered in this document.

ISA-88 is more than a standard for software, equipment, or procedures. It is a design philosophy. Understanding ISA-88 will help to better design processes and manufacture products.

How does ISA-88 help?

- Modularity allows for easier global replication and better return on investment.
- Isolates equipment from recipes.
- Design concepts make validation easier.
- ISA-88-aware solutions help track product and process data.
- Gathering requirements from customers and conveying requirements to vendors is easier.
- Provides guidelines on how to recover from abnormal events.

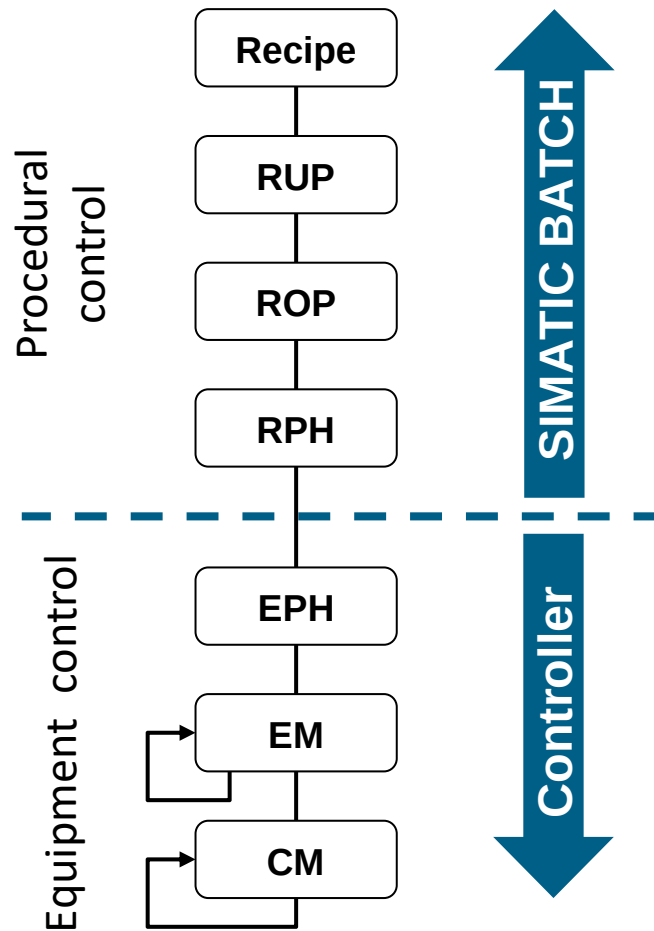
What is needed to define batch control?

Issue	Answer
How to make a product?	Recipes, procedures and formulars
What equipment is to be controlled?	Equipment entities
What control is needed?	Control activities

Therefore, three models and their relationships have been defined in the ISA-88 standard.

ISA-88 suggests that a batch management system, usually running on a computer, and an equipment control system, usually implemented on a controller, exist separately. In SIMATIC PCS 7 the recipe control is handled by SIMATIC BATCH, whereas the equipment control is handled on the AS410 controller.

Figure 2-1 Splitting of recipe control and equipment control



In the next chapters you will find explanation what each of these acronyms really mean.

3 Models of the ISA-88

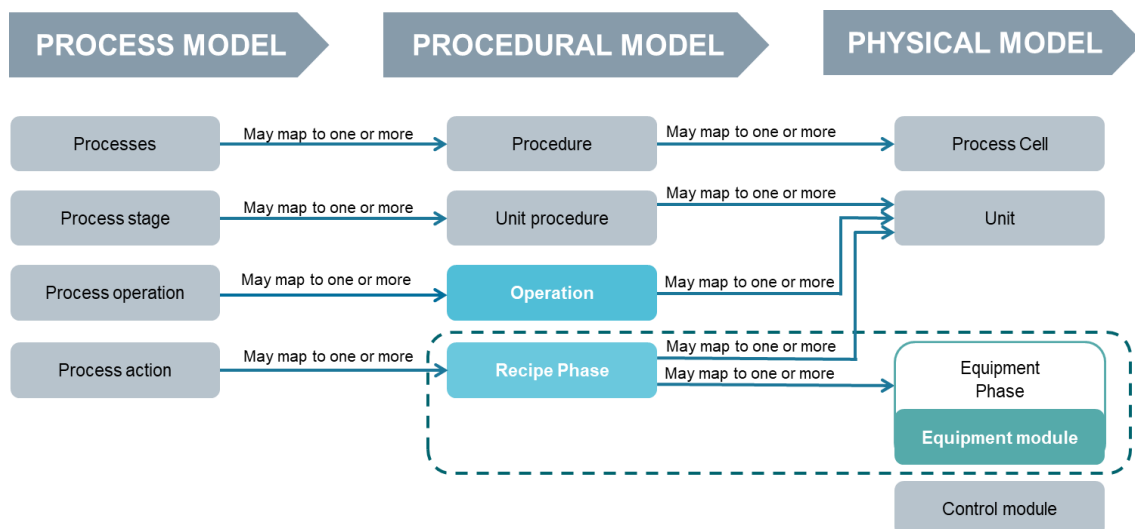
The ISA-88 defines three main models:

1. The process model, which handles the aspects of the chemical and physical changes during production.
2. The physical model, which handles the equipment being used during production.
3. The procedural control model, which handles the recipe structure for the product.

The relationship between the process model, the procedural control model and the physical model is shown in figure 3.1 below. The mapping of procedural control with individual equipment provides processing functionality described in the process model. The usage of equipment capabilities to accomplish desired processing tasks is a major point of this document. The equipment entity performs the functions defined in procedural elements. Procedural control elements are defined to correspond to elements of the process model to accomplish the desired processing for a batch. Procedural control elements are defined within a recipe.

Through the application of equipment control, the equipment entity will perform the functions defined by the recipe procedural element to accomplish a product specific task. This allows for a high degree of process flexibility that is essential in many batch processes. It allows for significant changes in process activities to be defined by a recipe author without reengineering equipment entities.

Figure 3-1 ISA-88 process/procedural control/physical models

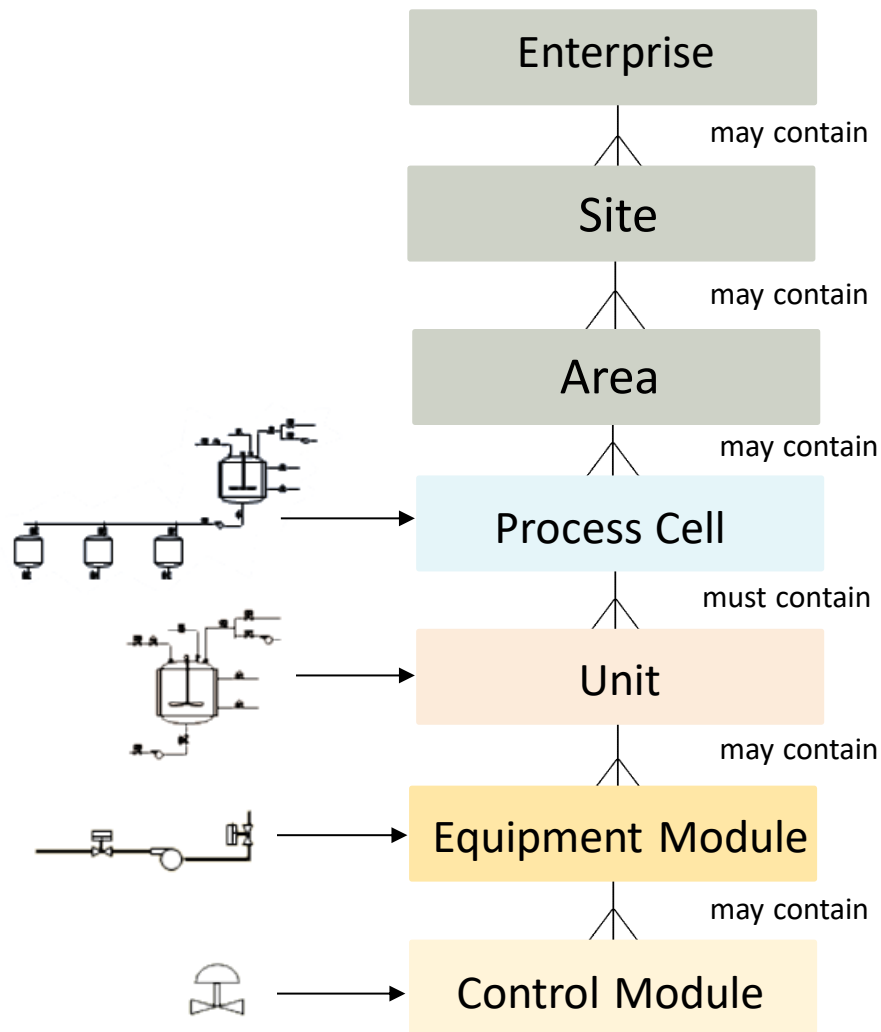


3.1 Physical Model

The physical model is described in detail in chapters 4 to 7, in this section only an overview is given. The physical model is shown in the following figure. The three top levels of the physical model are defined as Enterprise, Site and Area, but they are not relevant for this document and will not be considered.

This application example has been created with the following hardware and software components:

Figure 3-2 ISA-88 physical models



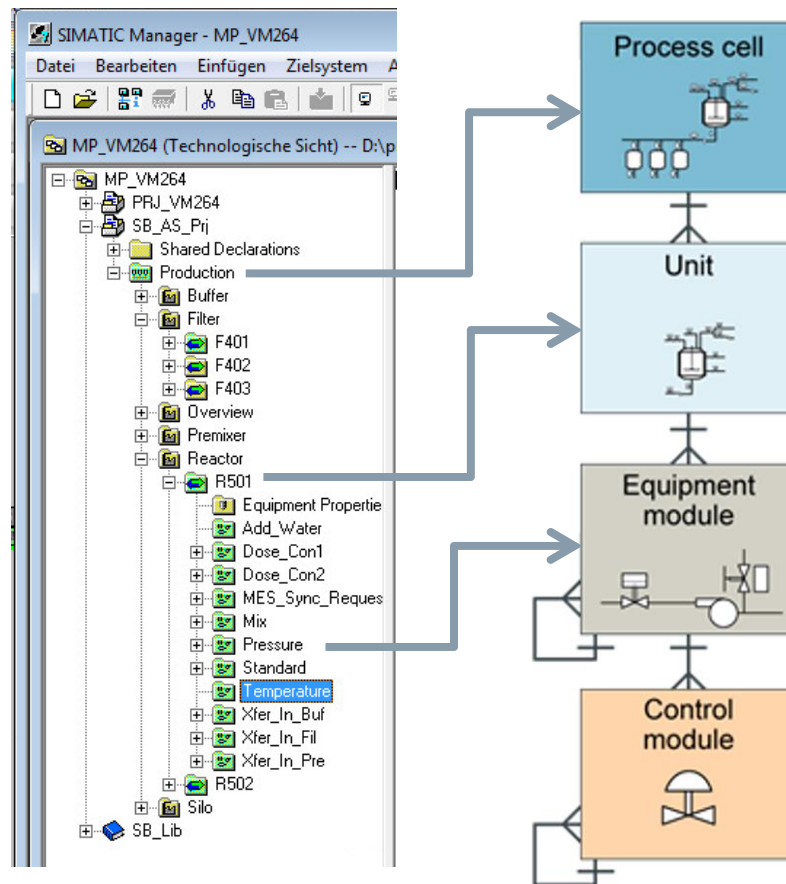
A process cell is a logical grouping of equipment (units, equipment modules, control modules) that includes the equipment required for production of one or more batches.

The actual batch process occurs within the units, such as a reactor, buffer or pre-mixer, of a process cell. During the production process only one batch can reside within a unit at a time.

A unit typically aggregates equipment modules, which in turn generally is implemented as an aggregation of control modules to fulfill certain tasks or process actions, such as dosing, agitating or temperature control. An equipment module is typically centered around a piece of process equipment such as vessel outlet/inlet manifold, process heater, a WFI drop point etc. Control modules represent single or multiple sensors or actuators or and are the actual processing equipment that perform one or more specific state-oriented or regulatory control tasks. Control modules are directly connected to the process through its I/O.

In this example it is shown how the physical model is mapped to the technological hierarchy in the PCS 7 Engineering System.

Figure 3-3 Mapping of the physical model to the PCS 7 technological hierarchy



3.2 Procedural Control Model

The procedural control model is a multi-level hierarchical model to accomplish the task of a complete process or part of a process based on a specific process cell. It is also the basis for defining recipe procedures. Each individual element in the model is a procedure, whose directly subordinate elements (if any) comprise its steps. The ultimate goal of the hierarchy is to efficiently organize the process-oriented tasks that are to be executed either automatically or manually at its lowest level.

The highest level of the procedural control model is the overall recipe procedure, commonly known as the recipe. The recipe can involve several production units, and the sequence of the processing steps of each of these units are captured in the recipe unit procedure. Within the recipe unit procedure consists of one or more recipe operations, which themselves contain subordinate recipe phases. A phase is the lowest level procedural element in the procedural control model which is either a representation in a recipe (Recipe Phase - RPH) or an implementation in equipment control (EPH) that is intended to accomplish all or part of a process action.

Phases can accomplish a process-oriented task, however, the set of steps that define its action are usually equipment specific. The scope of a phase is usually small enough that the specification of its actions (with appropriate variants, such as target values) is highly reusable from one procedure or set of equipment to another. When the full procedural control model is used, phases shall execute the required processing task(s).

Equipment phases coordinate and sequence specific process functions on a process unit. The exact detail of what each phase does, will be the subject of detailed design but the goal is to define phases that can be re-used in multiple recipes for different products. Hence there is a compromise between how much the phase should do and re-usability/flexibility. Common process operations such as charging, mixing, heating could all be incorporated into a single phase or these could be defined as separate phases, which could in theory be rearranged or even used standalone (by different recipe operations and recipe unit procedures that call them).

Examples of phases:

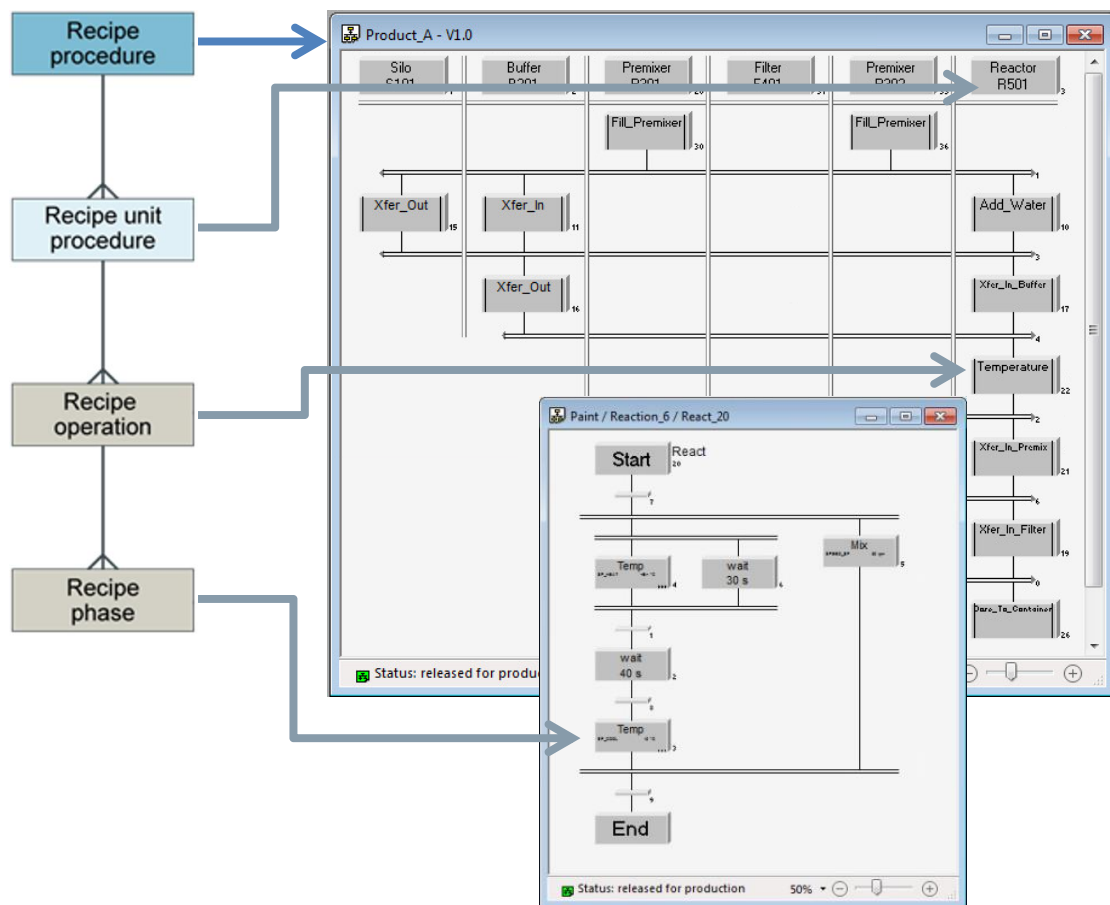
- Add vinyl chloride monomer
- Ramp and soak

The execution of a phase may result in one or more of the following actions:

- commands to basic control, such as
 - changing controller modes, initializing their outputs, and adjusting their set-points
 - setting, clearing, and changing alarm and other limits
 - modifying algorithm selections and tuning constants
- the collection of
 - process measurement values, such as tank level or liquid density
 - values calculated by basic control at the behest of the phase

In this example it is shown how the procedural control model is mapped to the different levels of a master recipe in the PCS 7 Recipe Editor.

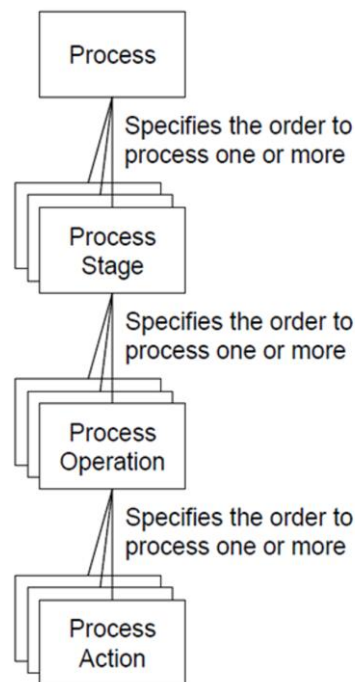
Figure 3-4 Mapping of the procedural control model to the PCS 7 master recipe



3.3 Process Model

In the ISA-88 standard the batch process is organized in the process model, which divides the entire process into a hierarchy of smaller subdivisions as shown in the following figure from ISA-88 (Reference \4\).

Figure 3-5 ISA-88 process model



Process Stages

Every batch process can be divided into process stages. Process stages can take place in series (after each other), in parallel (simultaneously) or a combination. Each process stage typically operates independently from other process stages. A sequence of physical or chemical changes to the material is usually the result of a process stage.

Examples: esterify, strip, neutralize, filter

Process Operations

Each process stage consists of process operations. They represent major processing activities, where the result is often physical or chemical changes to the material. Process operations are a bit more concrete than process stages.

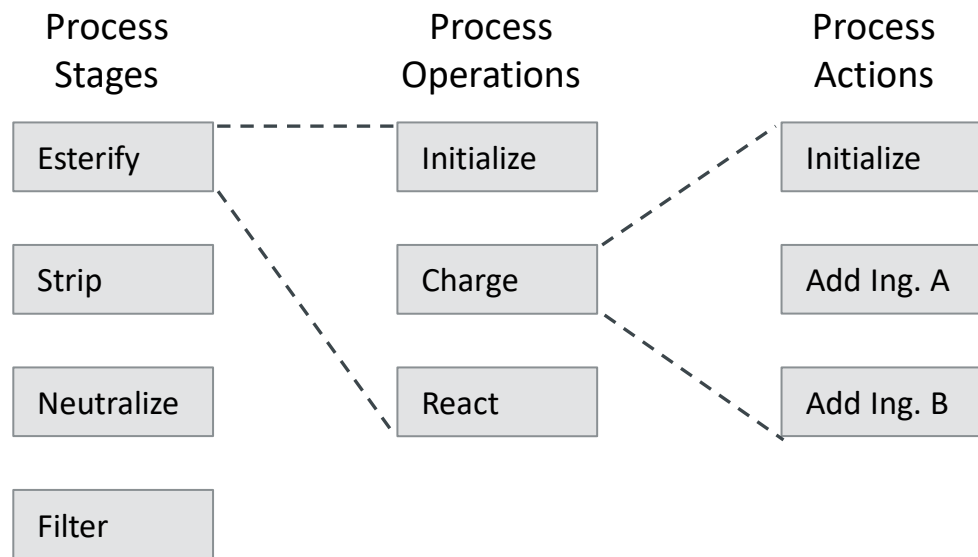
Examples: initialize, charge, react

Process Actions

Every process operation can be divided into process actions. They are the lowest level of the process model. The actions are needed to complete the operations.

Examples: add water, add other ingredients, heat the tank to 55 degrees, hold for 120 minutes

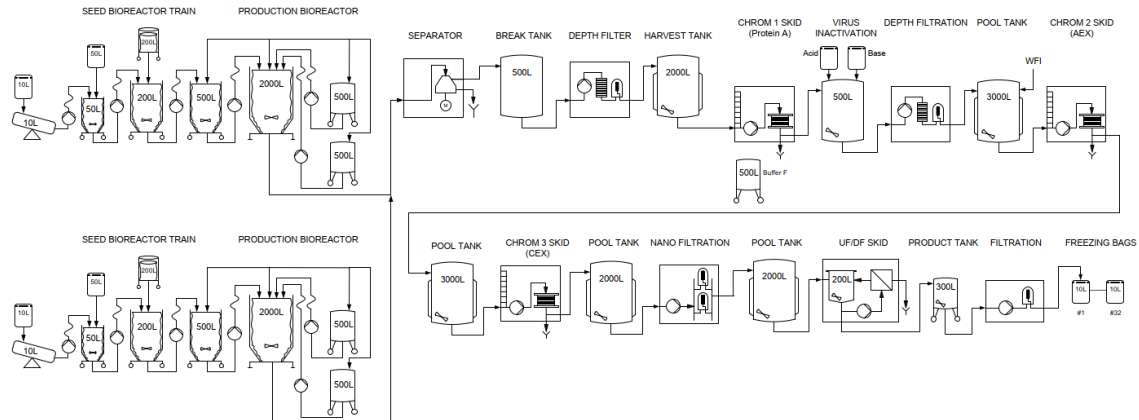
Figure 3-6 Example of a process model



4 From Process Flow Diagram to P&ID

In the conceptual design phase of a facility, the Process Flow Diagram (PFD) is established. This shows the general flow of plant processes and the equipment involved. The PFD displays the relationship between major equipment of a plant facility and does not show minor details such as piping, instrumentation and naming, a high-level example is shown in the figure below.

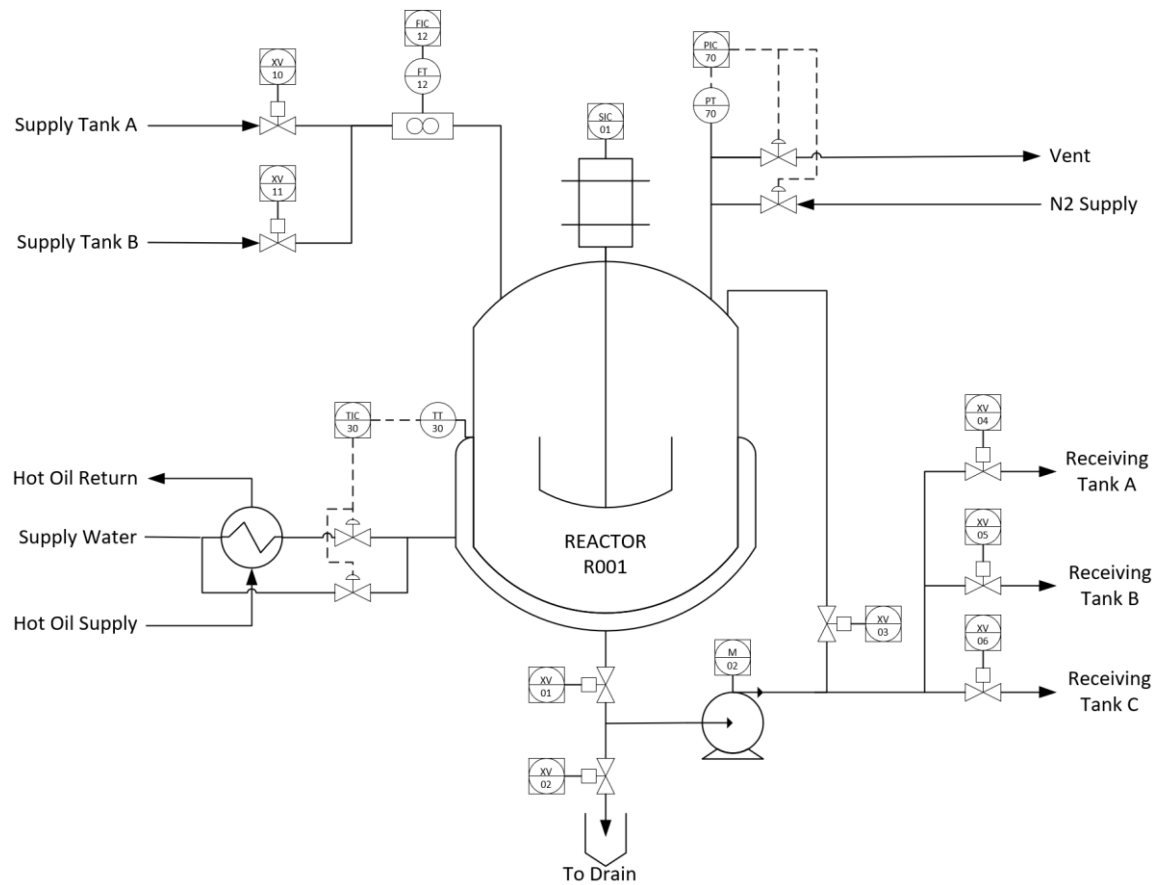
Figure 4-1 Example process flow diagram for a biopharma facility



In the basic design of a facility the PFD is augmented with more details and a piping and instrumentation diagram (P&ID) is established, which contains all the information needed for the automation of a plant, and for the structuring according to ISA-88. For the purpose of this document, a simplified P&ID is depicted in the next figure.

4 From Process Flow Diagram to P&ID

Figure 4-2 Simplified P&ID of a reactor vessel



The PFD and P&ID are closely linked to the ISA-88 physical model. Properly mapping elements of the physical model onto the P&ID will help us describe the functionality of the units and subsequently program it for automation. The next sections take a closer look at each of these elements.

5 Process Cell Classification

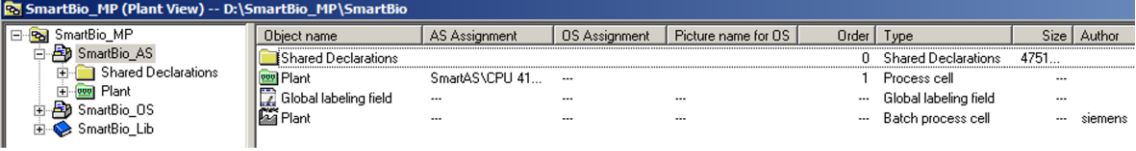
PCS 7 only reflects the equipment model of the ISA-88 physical model. As already mentioned, enterprise, site and area are not considered in this document.

A process cell, also abbreviated as PCell, contains all the units, equipment modules and control modules required to produce one complete batch. Normally the process cell encompasses the entire PFD and any supportive functions. A process cell may simultaneously run one or more batches.

A subdivision of a process cell is called an equipment train. An equipment train is composed of all units and other equipment that may be utilized by a specific batch. A batch does not always use all the equipment in a train and there may be more than one equipment trains within a PCell.

In PCS 7 the PCell is defined by the multiproject, as shown below. There can only be one PCell per multiproject.

Figure 5-1 Process cell in PCS 7 multiproject



Object name	AS Assignment	OS Assignment	Picture name for OS	Order	Type	Size	Author
Shared Declarations				0	Shared Declarations	4751...	
Plant	SmartAS\CPU 41...	---		1	Process cell	---	
Global labeling field	---	---	---	---	Global labeling field	---	
Plant	---	---	---	---	Batch process cell	---	siemens

6 Unit classification

The process cell is divided into a number of units. Units represent the physical processing equipment in the plant, such as a reactor, and are made up of equipment modules and control modules. A unit is an independent grouping of equipment usually centered around a major piece of processing equipment, such as a mixing tank or reactor, and it can perform one or more major processing activities — such as fermenting, harvest or purifying.

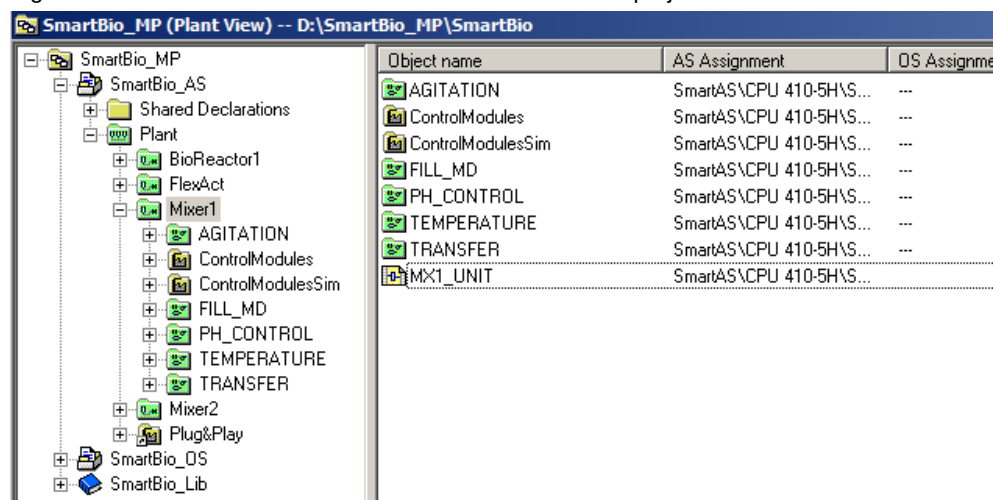
Usually, the modules that make up the unit cannot be manipulated or controlled by equipment outside of its own unit. Any equipment that is considered common or manipulated by multiple units should be defined as shared equipment, which can be temporarily acquired by a unit to carry out specific tasks, see also chapters [9.6 Shared EM](#) and [10.4 Shared Control Module](#).

An important consideration when defining batch units, is that a unit can only operate a single batch at the same time. The batch may exist in multiple units, but a unit can hold only one batch at any given time.

Defining the boundaries for units is important to guarantee the flexibility of the plant, reusability of the modules, and consistent reporting over multiple plants. The PFD can be used to initially identify the units. Once the P&ID details are known, the unit can be further refined. In case modular integration of intelligent unit operations, the unit and its ISA-88 structuring are defined by the OEM supplier.

In PCS 7, the unit and its modules are identified in the hierarchical folder structure of the plant view, as shown in below.

Figure 6-1 Unit “Mixer1” with its modules in the PCS 7 multiproject



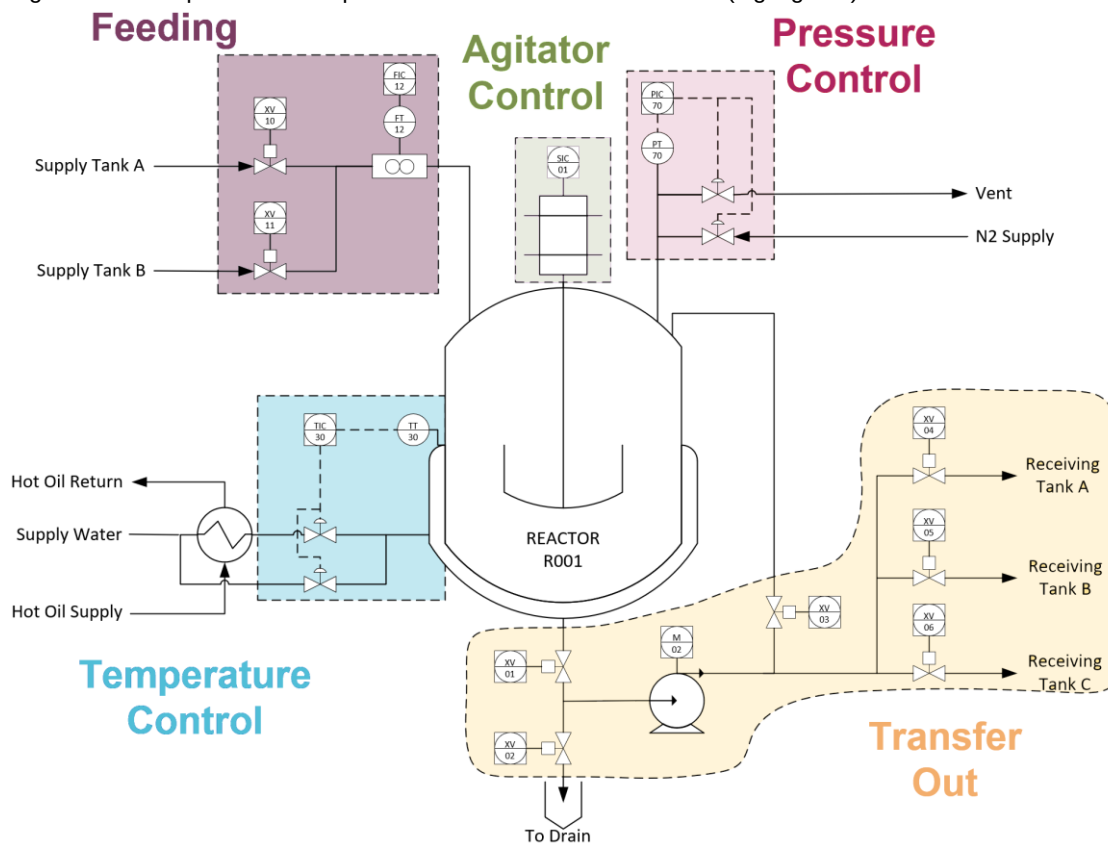
A recipe unit procedure is linked to possible unit candidates. The unit candidates determine which recipe phases can be used in the recipe unit procedure, and these recipe phases correspond to the common equipment phases for all unit candidates.

7 Control Module and Equipment Module classification

Once the unit boundary is defined, the lower level modules of the equipment model can be identified.

It's easiest to start with the **Control Modules (CM)**. These are the lowest level grouping of equipment in the physical model that can carry out basic control (state-oriented or regulatory control) and typically have a direct relationship to the type of instrument/device that it is being monitored and/or controlled. Examples are pumps, motors, valves, but also PID control loops. The CMs are typically directly connected to IO signals. CM also contain exception handling related to equipment, product and operator protection. The CMs are easily identified on the P&ID charts, as shown in the following figure. In PCS 7 the control modules can be controlled via the HMI by the operator in manual mode, or by the superior CM's or EM's in auto mode.

Figure 7-1 Example P&ID from process unit "Reactor" with its EM's (highlighted) and CM's



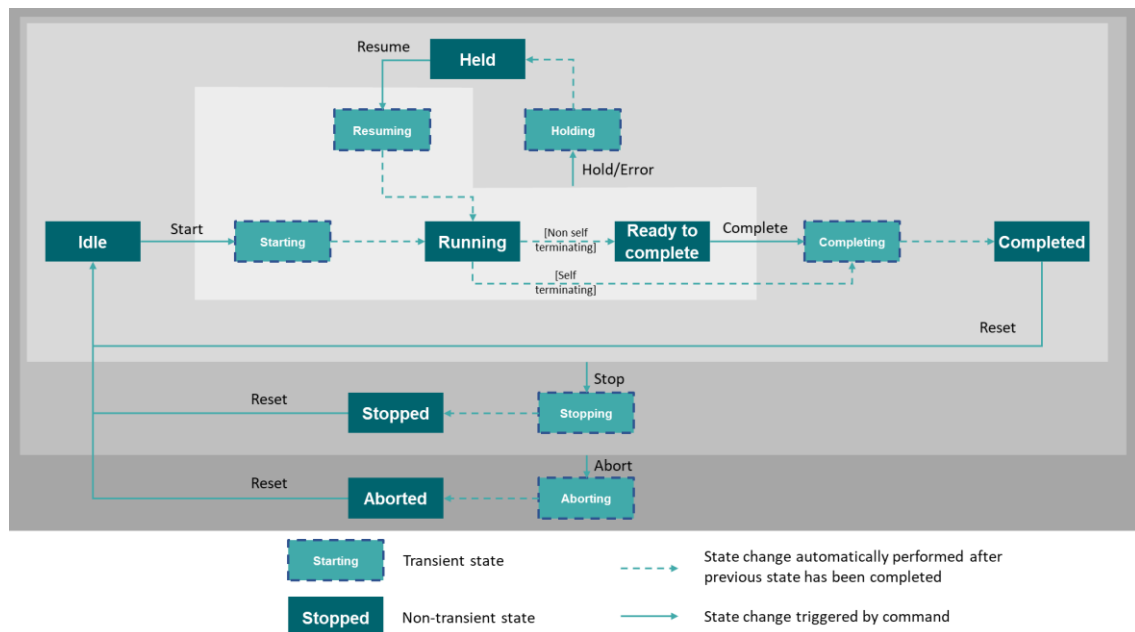
An equipment module is a functional group of control modules that can carry out a limited number of specific minor processing activities such as dosing, charging or temperature control. An equipment module is typically centered around a piece of process equipment such as vessel outlet/inlet manifold, process heater etc. The separation of a unit into functional **Equipment Modules (EM)** should already reflect the process actions and recipe phases from the process and procedural control model. Introducing a new product on the same equipment should be possible without altering the base sequences of the EM's. The EM's of our example reactor

are shown in the figure above: feeding, temperature control, pressure control, agitator control and transfer out.

According to ISA-88, it is possible for an equipment module to contain other subordinate equipment modules, although this is recommended to be restricted to areas where there are no phases or batch requirements.

An EM is usually a sequence in which all the subordinate CM's are monitored and controlled. The EM also contains exception handling logic that will handle failure of subordinate CM and process failure. Usually the exception logic is labelled as held state or held logic and drives the EM to a failsafe state. The EM itself does not have mandatory state logic defined in the ISA-88, although in PCS 7 an EM is usually recipe-aware and it therefore has a state logic that follows the state logic of the recipe phase, as shown in the next figure.

Figure 7-2 State logic of the procedural recipe phases



As the sequence of the EM reflects the process action, it is common to enable and disable control module related alarms and process interlocks or change alarm limits throughout the sequence. This must be considered within the design of such command sequences.

Arbitration of control modules shared between equipment modules within the same unit will be performed at the equipment module level, see also chapter [10.4 Shared Control Module](#).

Arbitration of equipment modules shared in the same unit or between units is performed at the phase level. Each phase will acquire the equipment modules needed to perform a specific process action. It is also possible to acquire equipment modules on another unit in the same way.

When partitioning the EMs within a unit, the following criteria should be applied:

- all CMs in the EM must support a similar processing purpose and be self-contained
- the EM must be reusable, so that a type-instance concept can be used
- the module has to be designed so that it can perform all of the available processing functions, see above
- EM must have the ability to operate on its own without other EM interactions

- ability to minimize the effects of process exceptions by containing them within the equipment module; ideally an exception will initially only affect one EM before being propagate to other EM's or even Units.

Like the CMS, in PCS 7 the equipment modules can be controlled by the operator in manual mode, or by the phases in automatic mode. The EMs linked to a particular unit are visualized in the PCS 7 hierarchical folder structure as subfolder of the unit folder, as was shown in [Figure 2-1](#).

It's important to note that the CMs and EMs should ideally follow the modularization approach where functionality is contained within modules that can be programmed and tested as standalone with a defined interface towards other modules. To increase reusability, within PCS 7 support for type-instance is made for the control and equipment modules. Similar EMs and CMs can be grouped in a type. This type can be tested and then all modules will be instantiated from this type.

By creating these standard ISA-88 batch elements, tremendous benefits can be achieved:

- savings in software designing and engineering
- faster testing and validation during the initial process of application implementation
- increased maintenance and flexibility - By means of the pre-tested standard types, the expansion of the installation can be realized in a faster way, with less (re-)validation effort.

8 Generic Equipment Module versus recipe-aware Equipment Module approach

There is a conceptual difference between the basic control functions of the physical elements and the procedural control elements. The functions of EMs and CMs are equipment oriented. You can for example start and stop the agitator of a reactor without having any material in the reactor. On the contrary, a procedural control element is product and process oriented. The logical separation of recipes and equipment provides the ability to separate product specific definitions, instructions and information (e.g. recipes) from processing capabilities (e.g. equipment entities). This separation of product information from production equipment capability enables the same equipment to be employed to make a number of products or have similar process equipment produce the same product with the same recipe.

There is an element known as the **Equipment Phase (EPH)** that links the physical model and the procedural control model. A recipe phase (RPH) in the control recipe references an equipment phase in the equipment control.

There are two classifications of equipment modules defined in the ISA-88:

- Recipe-aware equipment modules
- Generic equipment modules.

Recipe-aware equipment module

The recipe-aware equipment module is characterized by having one or more equipment phases that can be directly initiated by the execution of a control recipe. The EPH is limited to the EM interface (setpoints, state model, control strategies), which practically makes the EM directly visible in the recipe, hence the naming recipe-aware. This also means that the states, state transitions, and commands of the EM are limited to those recognized by recipes, and the EM state logic follows the RPH state logic as specified in the ISA-88 closely.

In SIMATIC PCS 7 the equipment phases of the recipe-aware equipment modules are represented by the interface between the EM and RPH, and contains the setpoints and the control strategies, as will be detailed in the next chapter.

The following figure shows the relationship between EM, EPH and RPH for the recipe-aware EM.

Figure 8-1 Linking between RPH and recipe-aware EM



Generic equipment module

The second type, the generic equipment module is very similar to the recipe-aware EM but lacks the ability to be initiated directly by the execution of a recipe. Equipment modules of this type have not direct relationship to recipes, but to procedural control elements like equipment phases and control modules. As such, their composition can be very flexible (the state logic does not need to follow the state logic of an RPH), and they may be commanded by other equipment modules, higher level equipment entities or an operator.

Maybe the desired EM type for this project?

In SIMATIC PCS 7 the generic equipment modules are triggered by the recipe via the equipment phases, which are sequences with a state logic that follows the RPH state logic as shown here.

Figure 8-2 Linking between RPH and generic EM



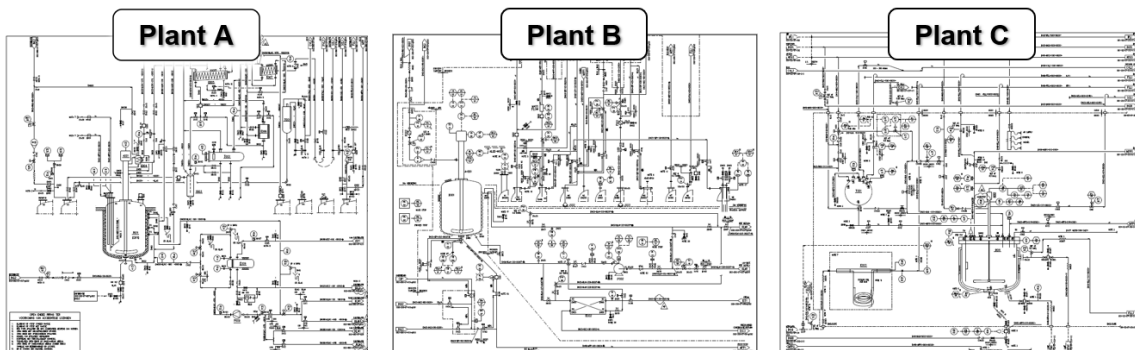
Why the need for the distinction between generic and recipe-aware equipment modules?

The top-level ISA-88 design pattern is the separation of product definition information from production equipment capability. This separation allows the same equipment to be used in different ways to make multiple products, or different equipment to be used to produce the same product. Product definition information is contained in recipes, and the production equipment capability is described using a hierarchical equipment model.

The separation of equipment capabilities – defined in an equipment hierarchy – and product manufacturing instructions – defined in recipes – provide the basis for improved flexible manufacturing. The master and control recipes' procedures contain recipe phases, and these are where the recipe is linked to equipment. Recipes contain references to equipment functions. The actual functions are performed by the equipment. The logic in a recipe's procedure specifies when the functions are to occur, in what order, and what the transition conditions are between functions. There is a substantial difference between general/site recipes and master/control recipes. The general and site recipes are equipment independent and describe the processing technique. Master and control recipes are specific to and dependent on the equipment in a defined process cell. They define the procedure that implements the process with actual physical resources.

Take the use-case where the same product is produced at various sites with the same general recipe. The equipment design is not necessarily the same between plants, as you can see in the following P&IDs, but the process actions to make the product are.

Figure 8-3 P&IDs of similar plants that can produce the same product but have different equipment



When using recipe-aware equipment modules, it could be that each plant needs a different master recipe for the same product, depending on the equipment specifics in the equipment modules, that have a direct link to the recipe.

The use of generic EMs and EPHs separate the product manufacturing instructions from the equipment capabilities to allow improved flexible manufacturing:

- Equipment phases: product process steps that call the various EMs (product specific)
- Equipment modules: minor process activity sequences that are equipment specific (plant specific) – For example temperature control EM can be done by electrical heating, adding steam or a manual operation. EM's of a certain class should have the same interface towards the EPH layer.

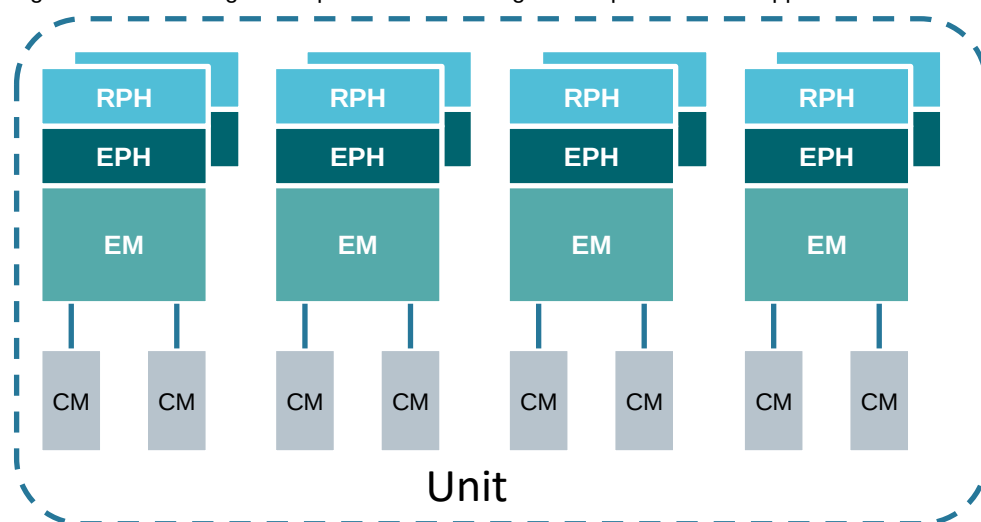
The process actions as defined in the process model, are minor activities, needed to make a certain product, for instance: temperature adjust, transfer, agitate, inert, blend. Process actions should be greatly standardized for re-use in multi-purpose / multi-product / multiple reactor plants or across sites. In case of generic EMs, the equipment phases are defined such that there is a one-to-one correspondence between a process action, a recipe phase and an equipment phase. All equipment dependencies are “hidden” from the recipe phase and recipe procedure.

9 The implementation of Equipment Modules in PCS 7

9.1 PCS 7 interpretation of EMs and EPHs

This section follows up with the more classical interpretation of EMs in SIMATIC PCS 7, the recipe-aware equipment modules:

Figure 9-1 ISA-88 logic for a particular unit using the recipe aware EM approach



Recipe-aware equipment modules

The equipment module consists of a Continuous Function Chart (CFC) in which a Sequential Function Chart Type (SFC type) is instantiated. Usually other logic such as interlocking, or a numerical computation is included in the CFC as well. The SFC type contains the state logic, which reflects the ISA-88 phase logic. Each state corresponds to a sequencer that is programmed according to the IEC61131 standard. It also contains the interface to the procedural batch engine, SIMATIC BATCH. This interface is known as the recipe phase on batch level, and the equipment phase at the controller level. This interface allows to:

- Choose the phase via the control strategy – more details on this below,
- Enter set points,
- Visualize actual process values,
- Visualize text information regarding actual step and acknowledgment of errors and graphically visualize the complete control strategy (sequence), transitions and actions in active steps.

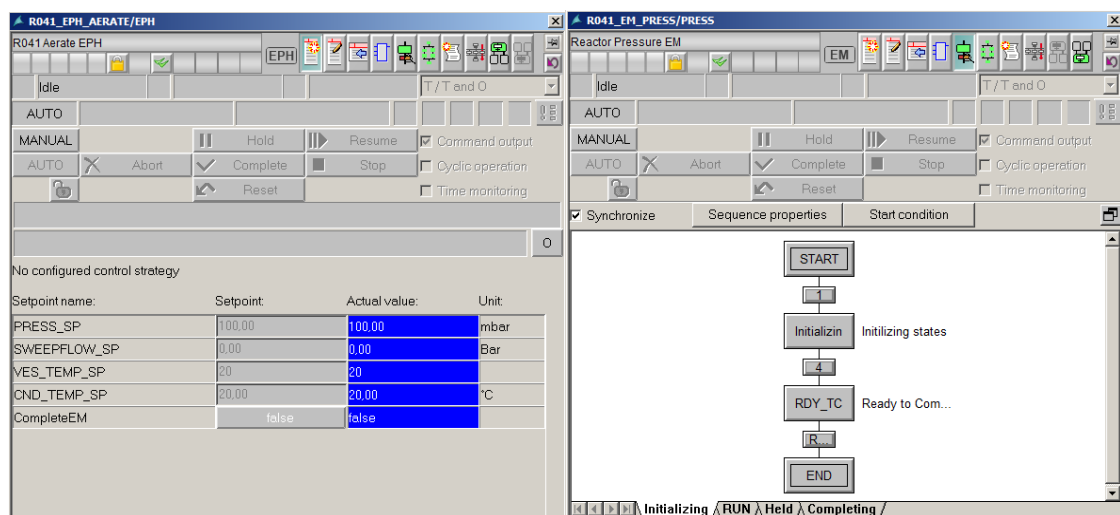
Each equipment module can have a number of different running modes or functions. Each of these is referred to as a "control strategy". For example, a reagent addition equipment module could have one control strategy for dosing a flow quantity, and another control strategy for dosing until fill level. Both control strategies execute the function dosing and use mostly the same control modules. Control strategies allow capsulation of several functions of the same control modules, without the need to add separate EMs for each function. Use of control strategies prevents complex sharing of control modules and makes the ISA-88 structure less complex.

Only one control strategy (phase) of an equipment module can run at a time. However, a switch from one control strategy to another without actually stopping the equipment module could be implemented if required. For example, a temperature control equipment module could have a control strategy to heat the jacket of the equipment to a certain set point and keep it there. It could have another control strategy to heat the product to a certain set point and keep it there. The first control strategy would involve a PID controller with the jacket temperature as measurement, while the second would involve a cascaded PID controller with the product temperature as the primary measurement and the jacket temperature as the secondary measurement. A switch from the first to the second control strategy when the product temperature reaches 90% of the product temperature set point could then be implemented in the recipe.

The EM controls and monitors its CMs, which are commonly directly linked as seen in the previous figure.

The sequence is visualized by a block icon and faceplate, which also shows the ISA-88 interpretation:

Figure 9-2 Faceplates for EPH and EM



A recipe-aware equipment module can be flagged as either an equipment phase or equipment operation (EOP), so it can be used on the recipe phase or recipe operation (ROP) level in the recipe.

Generic equipment module

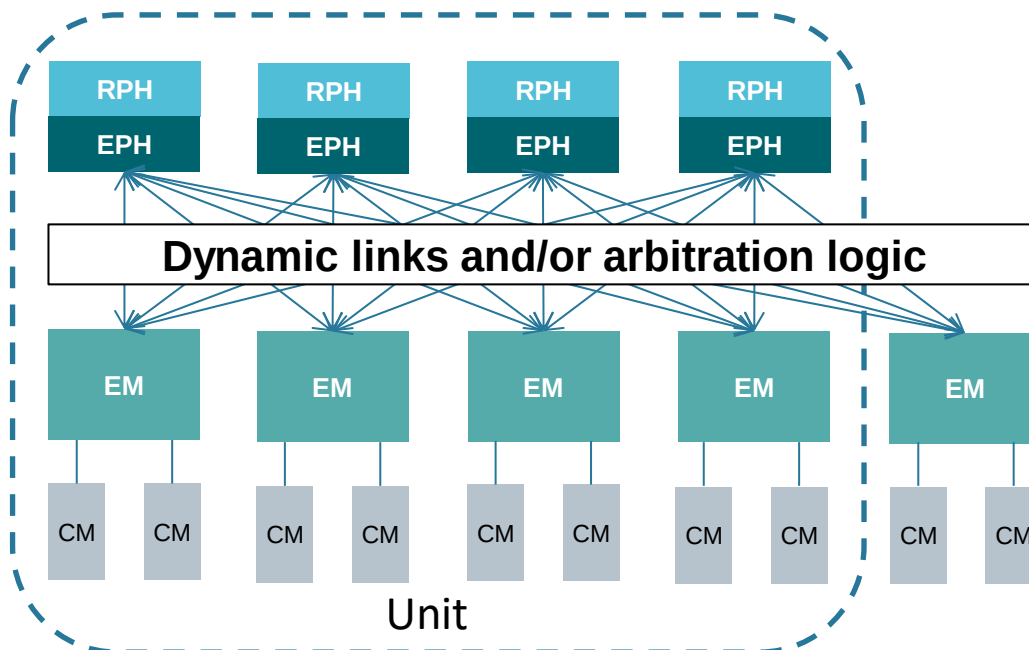
In the generic equipment module interpretation, an additional level of equipment phases is added to the ISA-88 logic in the PLC. These EPHs are very similar to the recipe-aware EMs and again consist of a CFC with a SFC type instance, in which each ISA-88 procedural state is reflected by a sequencer. The same properties as the recipe-aware EM, such as the control strategy, apply here as well.

The EPH level can either command EMs or CMs. Commonly, the generic EMs are programmed as SFC type sequences which still follow the SFC type state model. But since the EMs are not linked to procedural elements in this case, it does not need to follow the state model. Another interpretation at this level is the matrix approach, where an EM is a table of modes, and each mode controls control modules to a specified state.

Important here is that the equipment modules can be shared between several equipment phases, although with exclusive use – Only one EPH can control an EM

at the same time. Therefore, an explicit acquire and release of the EMs is needed. A multiplexer is built-in per default into the logic.

Figure 9-3 ISA-88 logic for a particular Unit using the generic EM approach



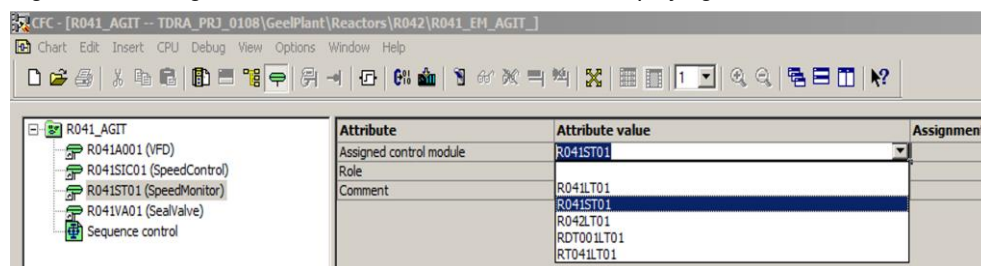
In both cases, the batch context, consisting of the batchname, the batchID and the recipe step number, is automatically sent at runtime from the batch to the equipment phase, and is from there propagated to the EMs and CMs.

9.2 What are EMTs and EPHTs?

Equipment module types (EMTs) and equipment phase types (EPHTs) are types for the EM instances and EPH instances. These types are centrally maintained in the master data library (MDL) of the PCS 7 multiproject. Instances can be created either manually in the project or automatically via the Plant Automation Accelerator (PAA). The main advantage between the type and instance is that changes can be synchronized from the type to the instance.

The underlying CMs and EMs are known as roles in the type, and upon instantiation need to be linked to CM instances and EM instances of the correct type, as shown in the next figure.

Figure 9-4 Linking the CM role to the CM instance when deploying EMTs



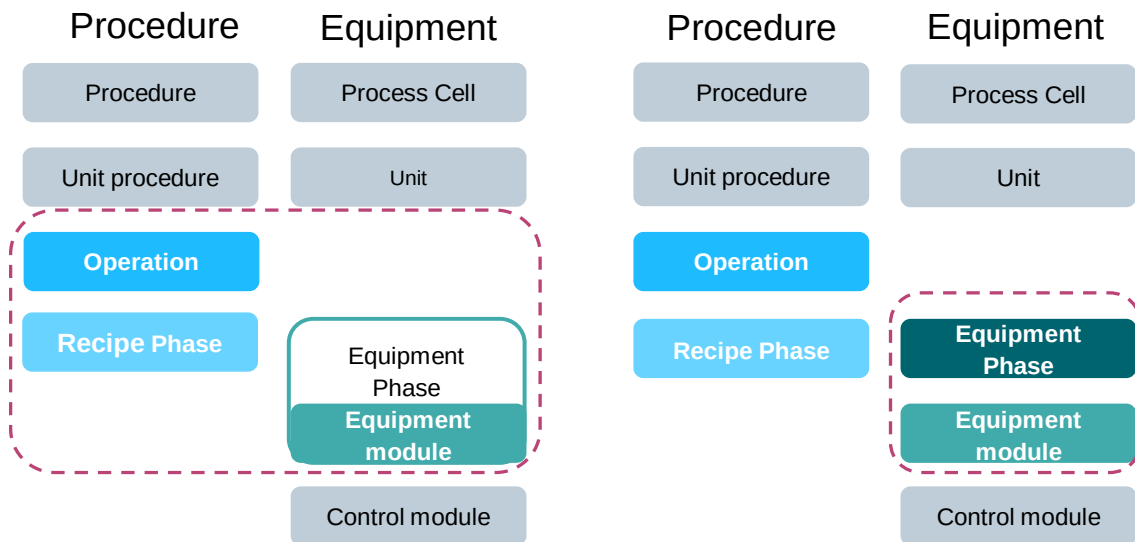
The type-instance concept promotes reusability of tested and documented types, over multiple Units, or even multiple PCS 7 projects or sites. Testing can be done virtual in combination with SIMIT Simulation Platform.

9.3 Design guidelines (comparison of both concepts)

Whether a plant should follow the recipe-aware or generic interpretation of an EM depends on the use case and plant philosophy. **In general, it's safe to say the choice is a trade-off between flexibility and portability.**

The figure below shows a comparison of the two approaches. There easily can be seen that in case of the generic EMs, the complexity of EM coordination gets shifted from the recipe operation level (at recipe level) to the EPH level (at controller level).

Figure 9-5 Visual comparison of the recipe-aware EM (left) and generic EM (right) approaches



The following table gives a summary of the benefits and deficits of both approaches:

Recipe-aware equipment module	Generic equipment module
Benefits	
- Flexibility for definition and change of master recipes by configuration of recipe operations - No revalidation of the EM's is needed in case of modification of recipe operations - Direct link between recipe phases and equipment modules - Change recipe procedures (ROP level) without changing controller programs - Change recipe parameters directly at EM-level - Reduced engineering effort for exception handling and assignment of EM, - as by default only one RPH can be active for an equipment module. - Current installed base with specifications and libraries of EM's available. - Takes advantage of technological engineering methods introduced with control module types	- No recipe dependency on process cell specific equipment - One central definition of master recipe for a product for global use. Easier to transfer recipes to different sites as recipe does not depend on process cell specific equipment - Less complex master recipes as complexity is shifted to equipment phase level - Coordination between equipment modules is easier at equipment phase level - Standardization: Reduced effort for (partly) re-use of equipment phases across different plants - Easier standardization at global level and more possibilities for digital support in development and life cycle - Possibility for parallel development of recipes (by process) and automation control (by automation) - Takes advantage of technological engineering methods introduced with equipment module types
Deficits	
- Dependency of recipe on process cell specific equipment - Multiple definitions of master recipes for same product in different plants (less standardization) - Makes it more difficult to exchange master recipes across different plants - Coordination between equipment modules is more difficult at recipe level	- More design effort needed by End user for ISA-88 conform control design (definition of EPH-level) - No current installed base with specifications and libraries of EM's available - Less details in master recipes by configuration of RPH at EPH-Level (EM coordination at controller level) - No direct link between recipe phases and equipment modules - Change recipe parameters indirectly at EM-level via EPH - More engineering effort needed for exception handling and assignment/monitoring/release of EM's for equipment phases - Revalidation of the EPH's in case of modifications (in the controller)

9.4 Principal approach

The Siemens Industry Online Support webpage is the gateway to find more information on correctly implementing ISA-88 in your PCS 7 project. Technical details on how to implement equipment modules and equipment phases are covered in SIMATIC PCS 7 Process Control System Compendium Part C:

<https://support.industry.siemens.com/cs/us/en/view/109761602>

A step-by-step description on how to practically implement the ISA-88 structure is given in:

<https://support.industry.siemens.com/cs/ww/en/view/109781909>

And an example project can be found here:

https://support.industry.siemens.com/cs/attachments/109781909/Batch_GS.zip?download=true

An online training is also available from Siemens Learning Campus:

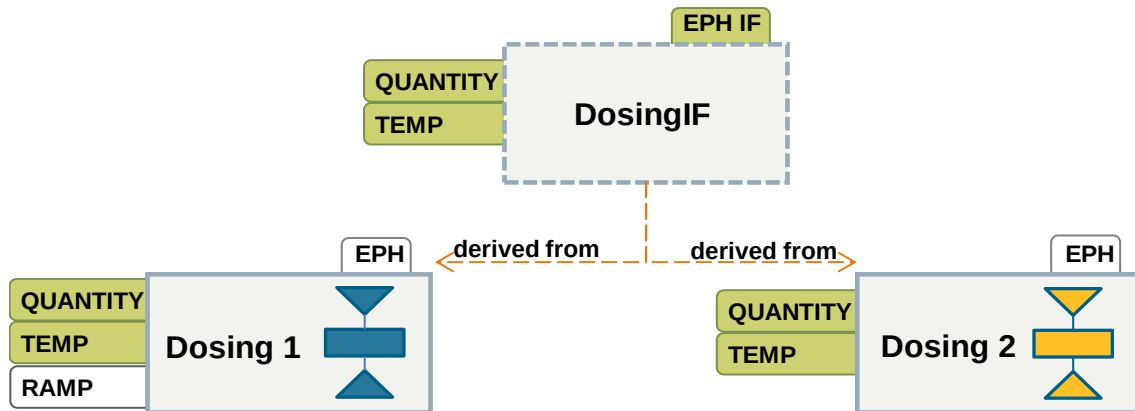
https://siemens-learning.sabacloud.com/Saba/Web_spf/EU2PRD0112/app/shared:spf-url=common%2Flearningeventdetail%2Fcurra0000000000003720

9.5 Derived EPH and EPH interface

It is possible to have several EPH types that have a very similar technological function, yet a different implementation on each unit in the same multiproject. For those EPH types, it is possible to define an EPH interface, so that the corresponding RPH in the recipe will be the same across multiple units. The actual content of the sequencer can then differ between the various derived EPHTs.

First an EPH interface object is constructed, which contains the interface to SIMATIC BATCH, but no sequence contents. Next the actual EPHT is derived from the interface and can be extended with additional parameters and sequence logic, as depicted in the figure below. Note that the derived EPHT can have more parameters than its linked interface, but those additional parameters are not visible on batch level, as they are not included in the interface. They can thus only be manipulated on the HMI faceplate.

Figure 9-6 Defining the EPH interface (IF) and the derived EPHs



9.6 Shared EM

In the case of generic equipment modules, the EPH can easily acquire EMs beyond its unit boundary, see [Figure 9-3](#). The built-in multiplexer takes care of the arbitration of the shared EM.

10 The implementation of Control Modules in PCS 7

10.1 PCS 7 interpretation of Control Modules

A control module is typically a collection of sensors, actuators, other control modules, and associated processing equipment that, from the point of view of control, is operated as a single entity.

Examples of control modules are

- a regulating device consisting of a transmitter, a controller, and a control valve that is operated via the set point of the device.
- a state-oriented device that consists of an on/off automatic block valve with position feedback switches, that is operated via the set point of the device,
- a header that contains several on/off automatic block valves and that coordinates the valves to direct flow to one or several destinations based upon the set point directed to the header control module.

The control module types (CMT) in PCS 7 enable a comprehensive type-instance concept and thus highly efficient handling of types and instances over an entire project.

CMT are always created and updated in the master data library. They contain all information necessary for the function of the measurement loop (function blocks, signals, parameters, messages ...). The instances are generated from the CMTs, either in PCS 7 engineering tools themselves or with COMOS/PAA.

CMTs essentially consist of a conventional CFC with the function blocks and the connection logic as well as the technological IOs. These technological IOs:

- contain the commands and statuses for the use in the above-mentioned equipment module type,
- enable the use of variants,
- are the interface to COMOS.

Due to the split into CFC and technological IOs, the CFC part can be edited as before and the CMT as a whole can be handled from a technological point of view. Technological objects are created in the technological IOs and connected to the blocks, signals, parameters and messages in the CFC part. This means that the technological connections can be designed according to technological aspects, independent of the function modules used.

In addition to the mentioned blocks, signals, parameters and messages, standardized commands and statuses can be created in the types which considerably facilitate the project engineering of the sequences in the equipment module type.

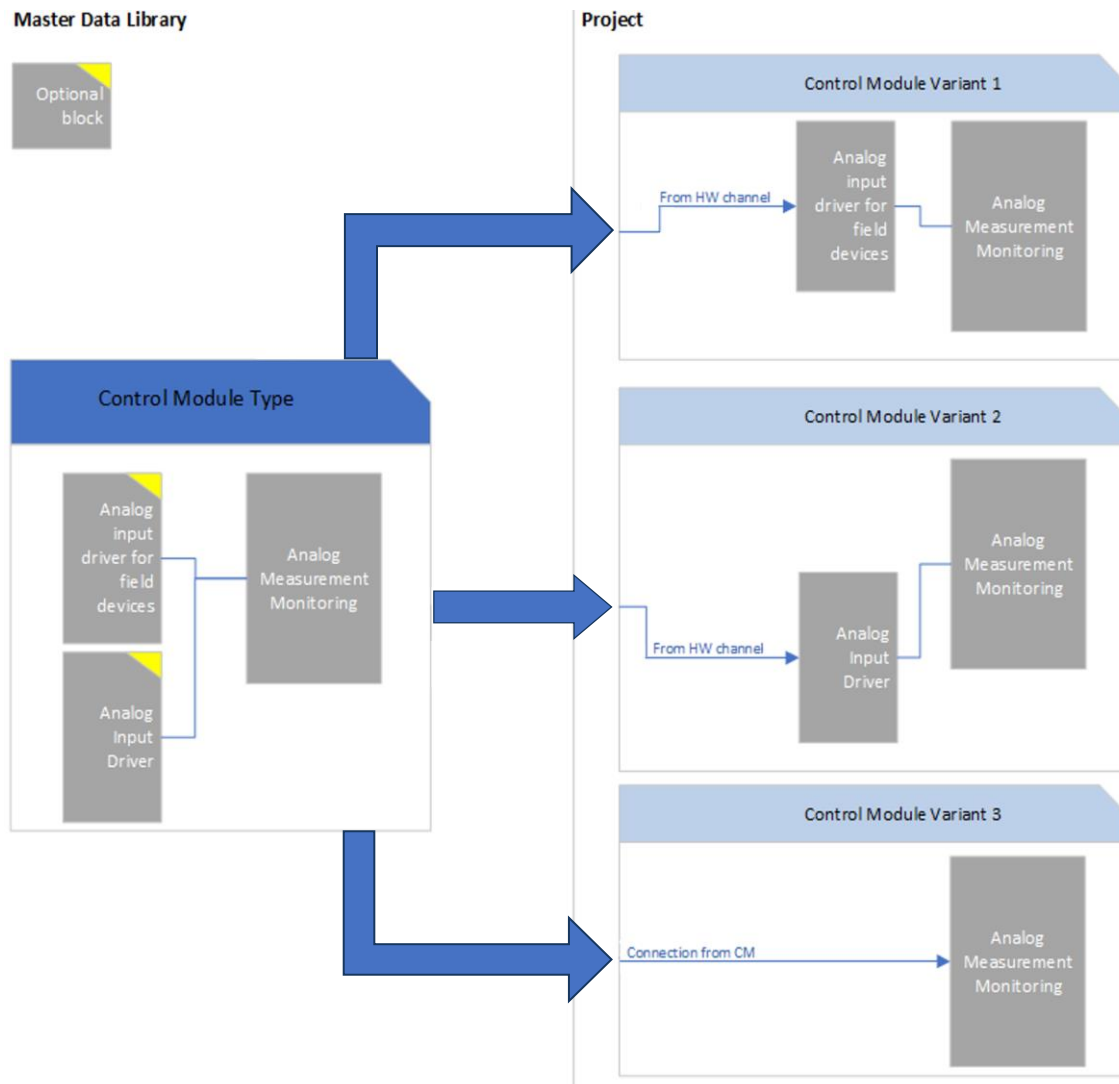
The decisive advantages of CMT compared to conventional technology are synchronization and variant creation.

By using CMT, it is possible to synchronize subsequent changes of types, such as connection changes or additional blocks, into the instances without having to create the instances again or losing already done detail engineering. Instance-specific changes are protected against unwanted changes. In addition, a detailed list of all changes to be executed is provided before the actual synchronization.

If instance-specific differences, such as additional interlocks or different driver blocks are required, there are two basic ways to implement them. Which one to choose depends on the type of difference.

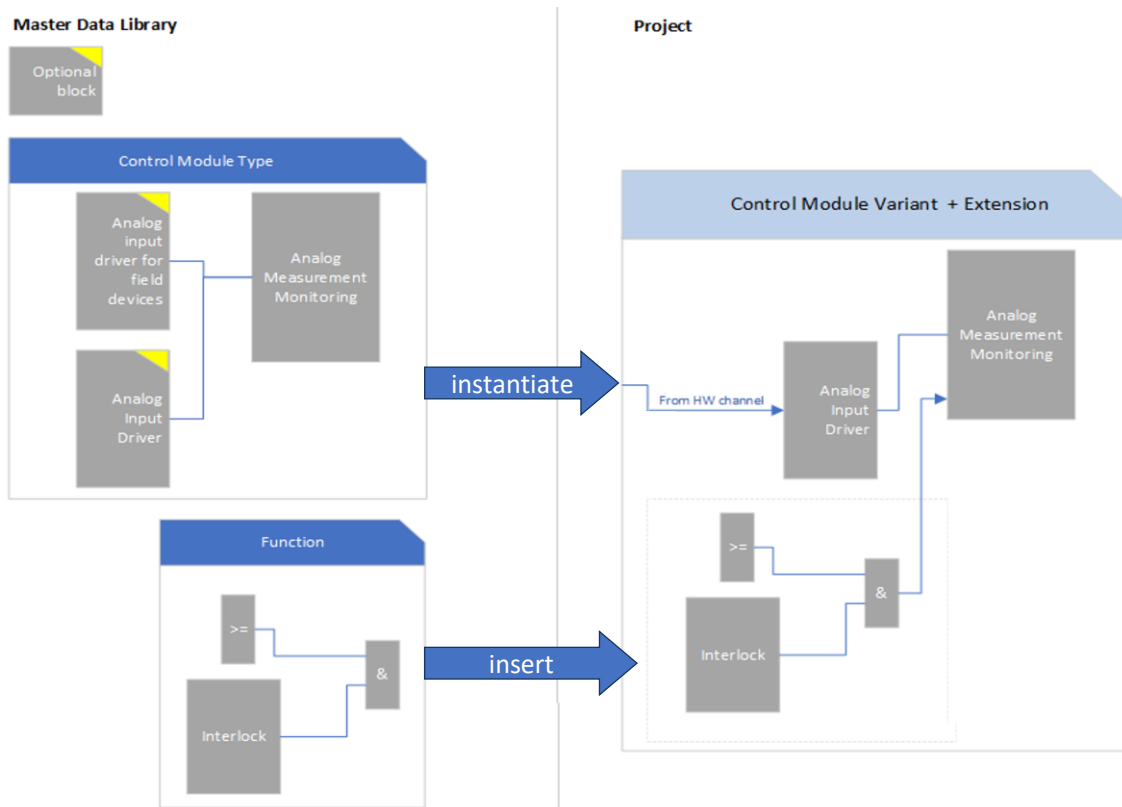
The type offers a choice between instance-specific functions. For example, should different channel drivers with the same type functionality be used, or functionally identical valves with or without feedback be implemented? In these cases, the “optional” functionality is used in the technological connections of the CMT. A change from one variant to another is possible at any time.

Figure 10-1 Basic principle of variants in CMT



To insert a function extension into one or more instances without changing the type, independent functions can be created in the master data library and integrated into the instances. These functions are always handled separately from the CMT. As with the synchronization of the CMT, it is also possible to make subsequent changes to the functions and synchronize them in the projects.

Figure 10-2 Basic principle of functions in a CMT



10.2 Benefit

The highest priority must be given to the type-instance concept. Quick changes in an instance would be a violation of the concept and would, for example, lead to risks in synchronization.

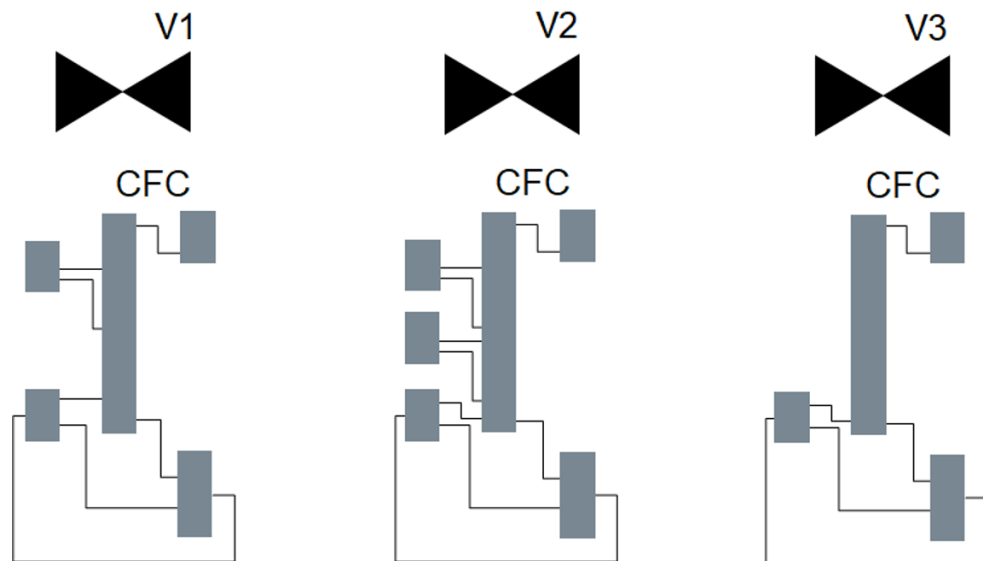
In contrast to this limitation, the type-instance concept offers many advantages:

- Adaptation of changes from type to instance without unnecessary modification of the existing application is possible.
- Changes in the central/general area only need to be tested once.
- System-side proof that the functions are the same in all instances is given.
- A type can contain several variants.
- Change from one variant to another variant without losing connections already created.

10.3 Principal approach

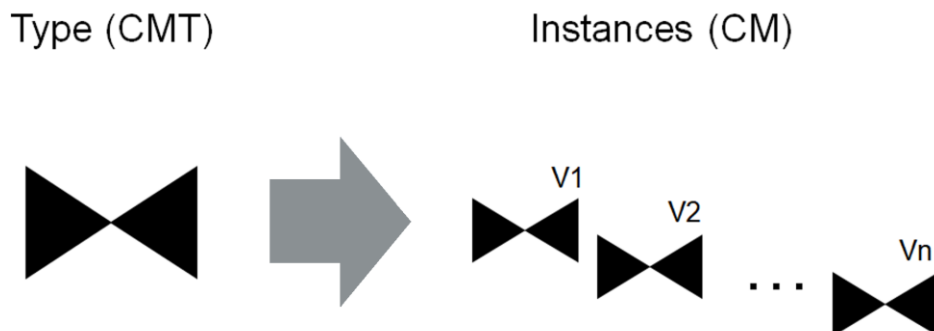
The basic idea of the type-instance concept is to create a type from similar functionalities, which covers all possible features. For example, when creating a valve type, it must be considered whether feedback signals are available. Is a feedback necessary? How is the valve controlled? Which interlocks are required? All these functionalities have to be considered when creating a type and finally implemented with fixed and/or optional function blocks.

Figure 10-3 Basic principle with CFC



Once the type design is completed, the instances can be created.

Figure 10-4 Basic principle with CMT



Further information can be found in the following application examples and FAQs:

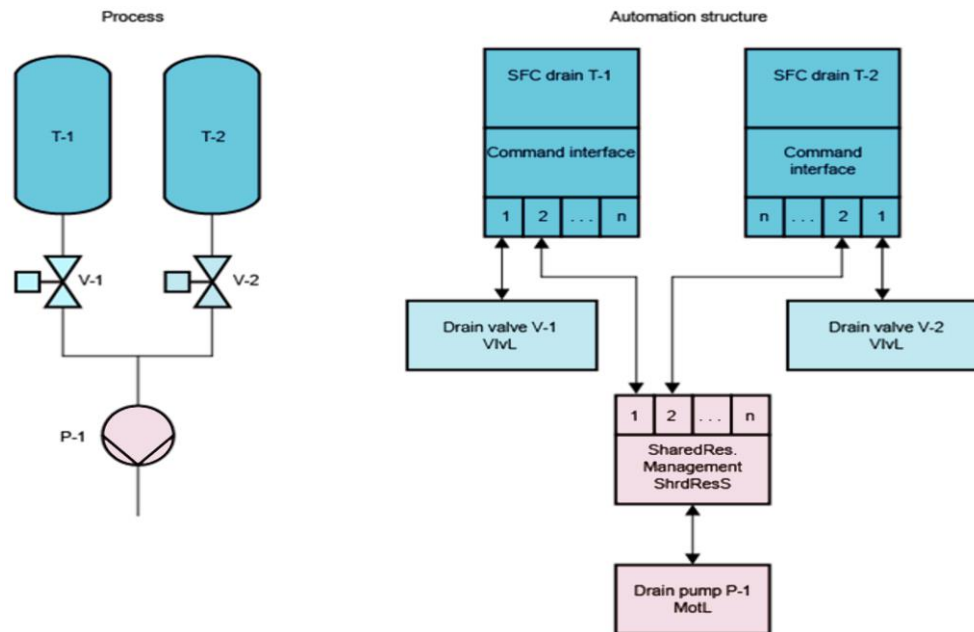
- FAQ 109475748:
[Control Module \(CM\) Technology – Efficient Engineering with SIMATIC PCS 7](#)
- FAQ 109758382:
[Synchronization of individual control module types \(PCS 7 V9.0 SP1\)](#)

10.4 Shared Control Module

In case the same equipment e.g., a motor should be used by several SFC, there is a function block that organizes individual accesses from upstream (preceding) applications to a downstream (subsequent) block.

The block allows to coordinate the access from up to four SFC step sequencers to a technologic block of the families "Drives" or "Dose".

Figure 10-5 Shared control module



The block can be cascaded via the fourth channel so that accesses via more than four upstream applications are possible.

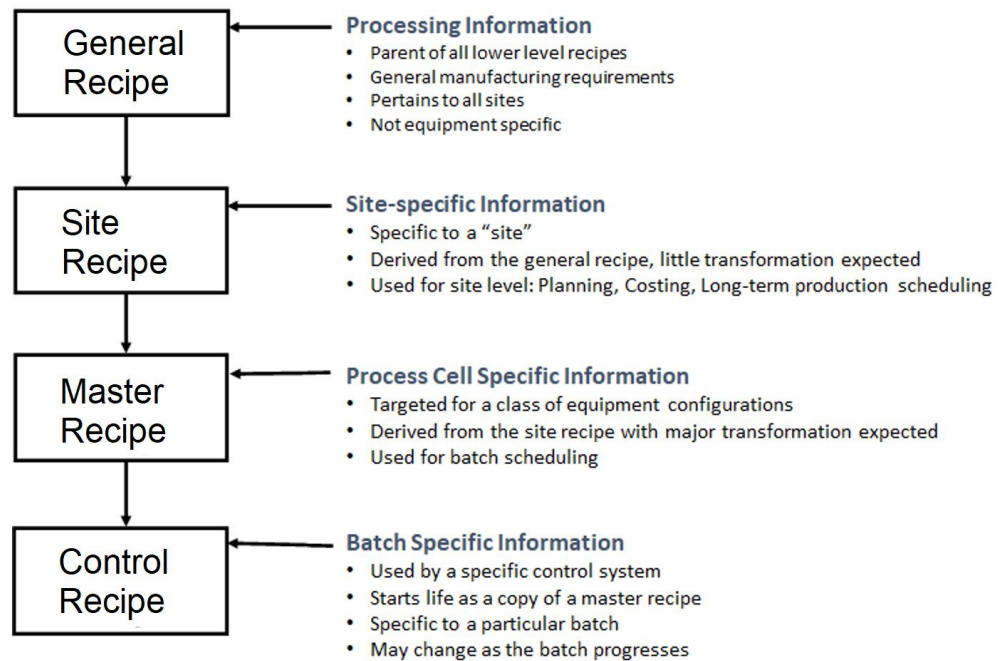
More detailed information can be found in „SIMATIC Process Control System PCS 7 Advanced Process Library (V9.0 SP3) Chapter 7.7.

11 Implementation of the Procedural Control Model with Recipes

How to make a product is defined in recipes.

ISA-88 defines four different recipe types as follows:

Figure 11-1 Recipe types



General recipes and site recipes are not developed in PCS 7.

The Recipe Editor of SIMATIC BATCH provides a comprehensive set of functionalities to create master recipes and develop structures to challenge difficult process situations. With the procedural and structural elements, it is very easy and intuitive to create sequences to control the batch production process. In case automated exception handling is needed, monitoring areas and command steps can support operators during critical situations.

The control recipes are derived from the master recipe and can be monitored and controlled at runtime within the BATCH Control Center or through controls embedded in the PCS 7 OS process screens.

These components and features are described in the sections below.

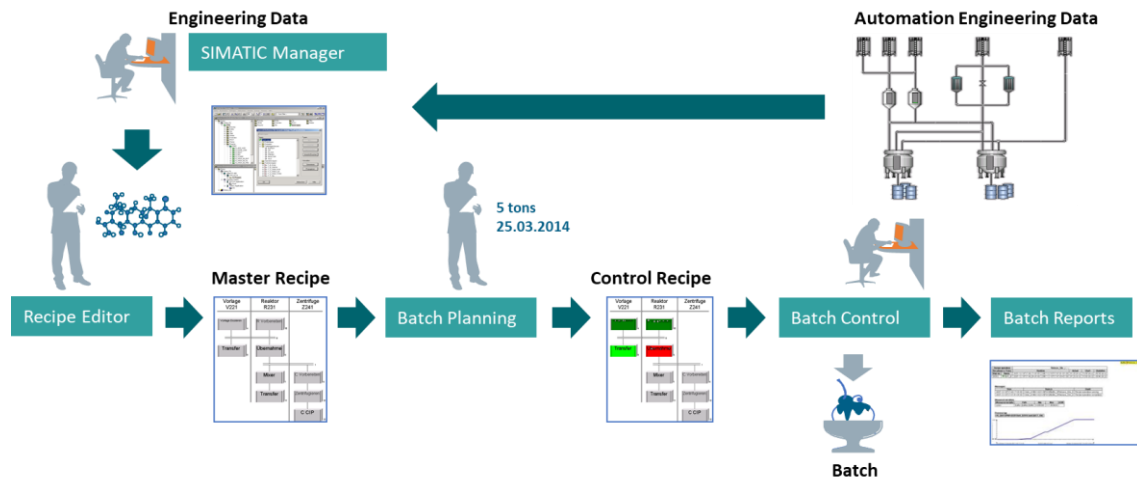
11.1 Batch equipment engineering

During a process analysis the plant is structured into a process cell, units, equipment modules and control modules. This task eventually results in the technological hierarchy of the PCS 7 engineering system.

Based on this equipment model (ES technological hierarchy) the procedural control model (master recipes) can be created. No double engineering of batch-specific engineering data is required, and no additional equipment editors are needed.

At runtime the control recipes will be derived from the master recipes to produce the batches.

Figure 11-2 Engineering workflow



11.2 Recipe engineering

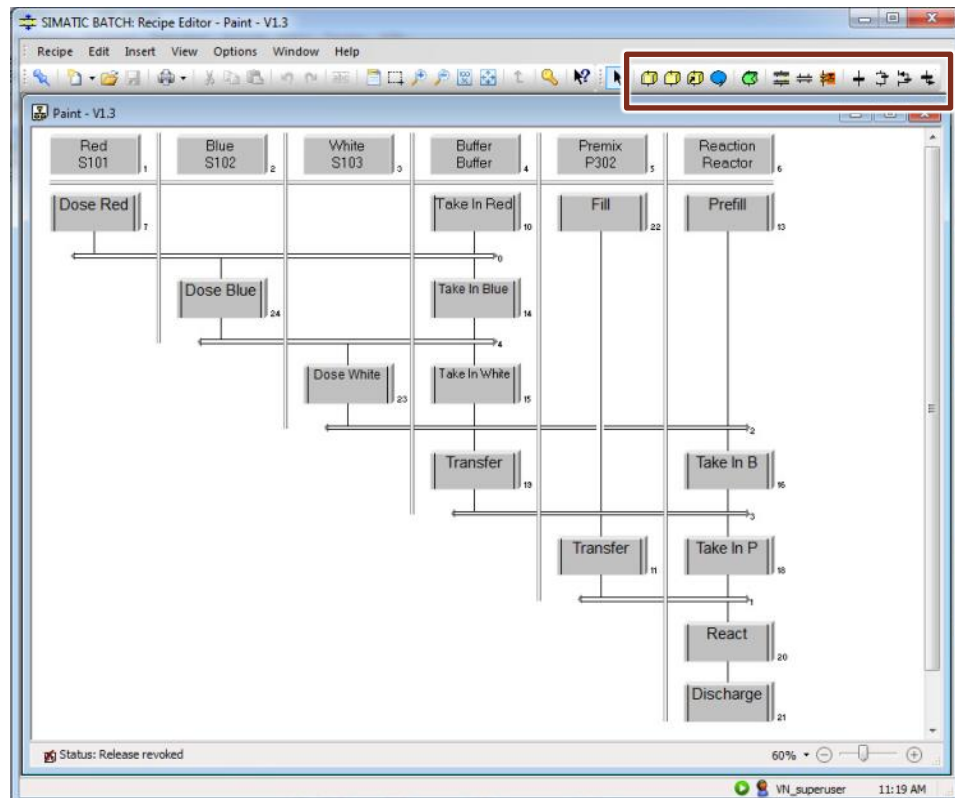
PCS 7 SIMATIC BATCH provides a very powerful recipe editor with which the procedural control model can be implemented in a convenient manner.

Recipe editor supports

- creation and modification of master recipes and library objects, so that it is "engineering instead of programming",
- documentation of master recipes and library objects,
- plausibility check and
- release of master recipes and library objects for test or production.

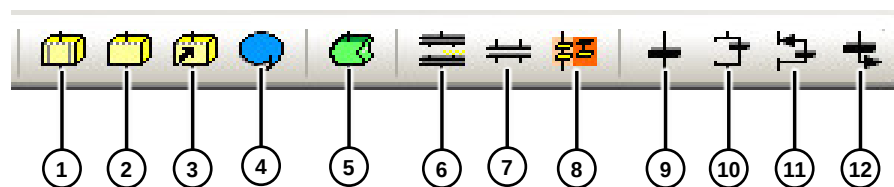
11 Implementation of the Procedural Control Model with Recipes

Figure 11-3 Example for recipe engineering



In detail PCS 7 SIMATIC BATCH provides the following functions for engineering of recipes:

Figure 11-4 Functions in PCS 7 SIMATIC BATCH for recipe engineering

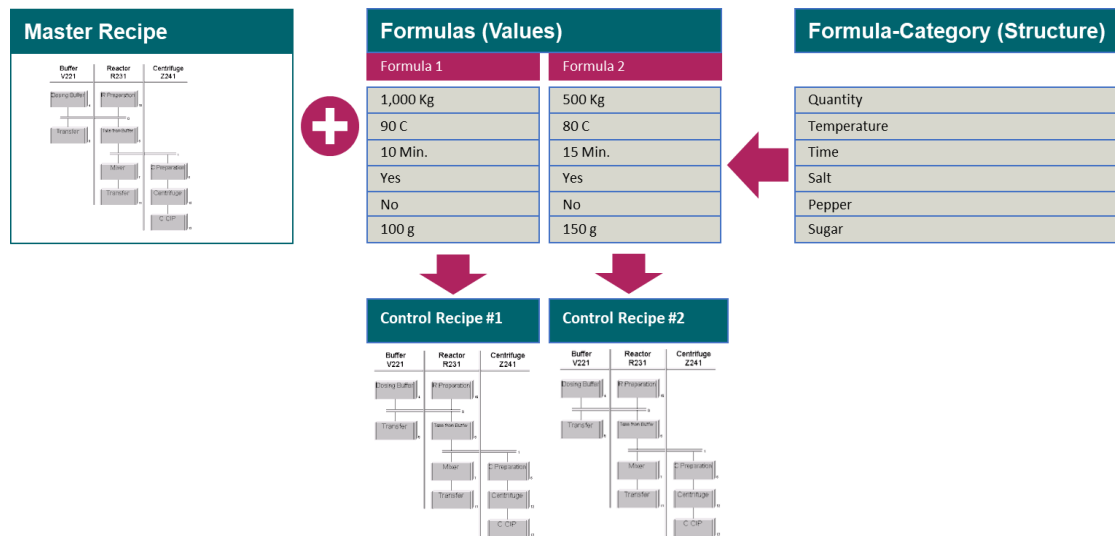


- | | | |
|-------------------|-------------------|----------------------|
| ① ROP | ⑥ Parallel branch | ⑨ Transition |
| ② RPH | ⑦ Sync. line | ⑩ Alternative branch |
| ③ ROP library | ⑧ Monitoring area | ⑪ Loop |
| ④ Operator dialog | | ⑫ Jump |
| ⑤ Command step | | |

11.3 Separation of procedure and formula

With one master recipe combined with several formulas, many control recipes (products) can be created. This makes it very easy to modify the production process without changing many recipe procedures for similar products.

Figure 11-5 Formula concept



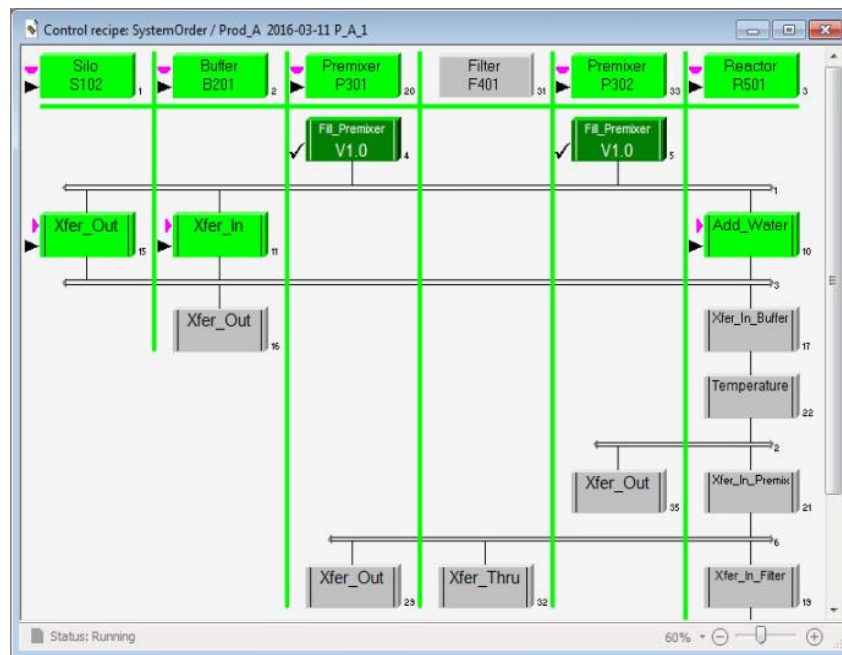
11.4 Coordination of Units

Most batch production plants contain several units which have to be synchronized at specific points during the production process, e.g. when a transfer takes place from one unit to another.

SIMATIC BATCH provides a convenient solution for this synchronization called synchronization line. No additional efforts are required on the engineering level. The graphical representation is very clearly laid out during recipe creation and at runtime for the operator.

There is also an option to block the progress within each recipe unit procedure (RUP).

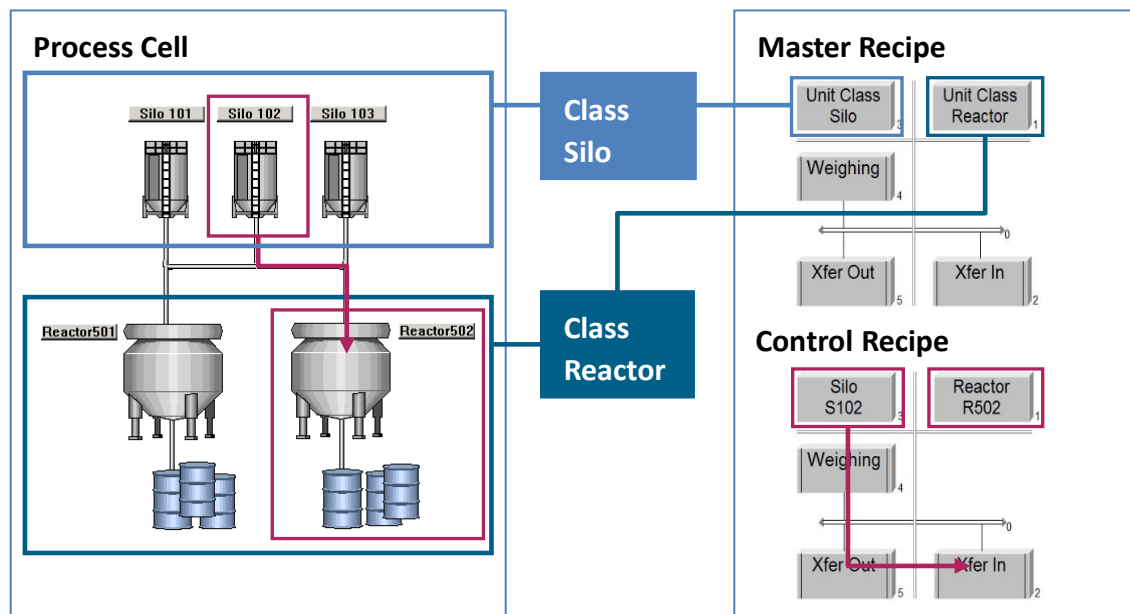
Figure11-6 Example for synchronization lines



11.5 Late binding strategies

With SIMATIC BATCH unit class-based recipes can be created. The class concept allows equipment independent recipes (separation between recipe and equipment) for facilities with similar units (unit classes) for optimized utilization. During recipe creation it is not necessary to select a specific unit for a recipe unit procedure, instead a list of units that are candidates to execute the RUP is created. With the late binding concept, unit allocation takes place at runtime, based on the allocation strategy and unit constraints.

Figure 11-7 Class-based recipe concept



SIMATIC BATCH provides four different unit allocation strategies, each RUP is configured to use a specific allocation strategy to select a unit from the list of candidates.

- Operator selection: When the RUP is activated the operator can select an appropriate unit candidate.
- Preferred unit: A unit candidate will be selected during recipe creation in the master recipe.
- The longest out of use: A unit candidate that hasn't been used for the longest amount of time will be allocated.
- Process parameter: A unit candidate can be specified by an external system or can be selected by controller logic and be deferred up to a process parameter.

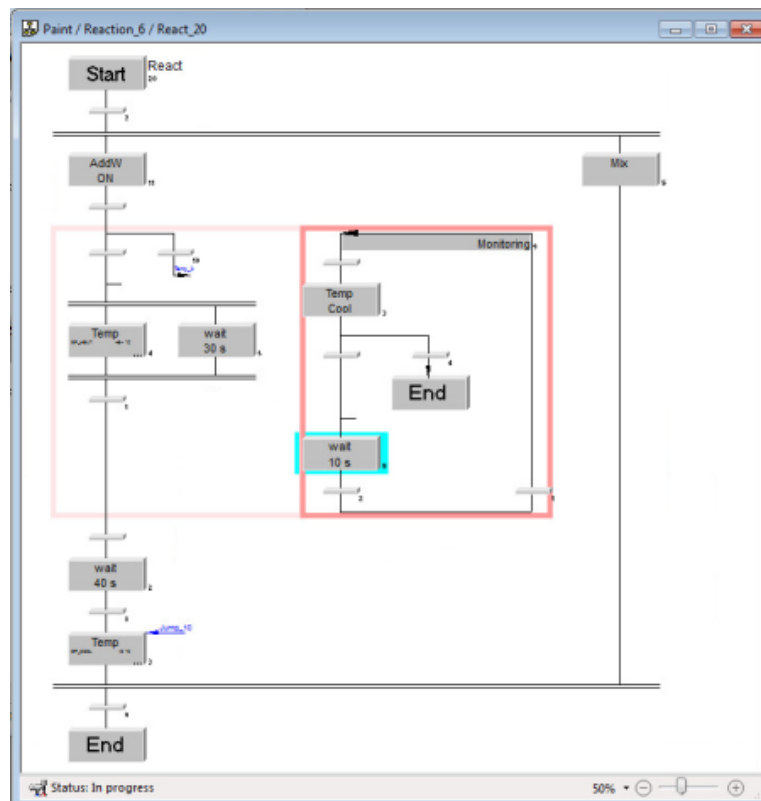
The selection of a candidate unit is defined by one or more conditions. Static and dynamic attributes can be evaluated. Static attributes can be equipment properties like e.g. "vessel size", "max. temperature", "unit class". Dynamic attributes can be process parameters like "current pressure inside a vessel", "actual temperature of the product".

11.6 Monitoring area

For exception handling process events within the procedural model can be monitored as well as a defined area of a recipe procedure (monitoring area).

A transition condition at the start monitors process values, states and events. When the transition condition is fulfilled, the monitoring runs concurrently. Monitoring is configured similar to all recipe structures and can utilize all the topological elements (loops, AND, OR, synchronization lines, etc.).

Figure 11-8 Example monitoring area

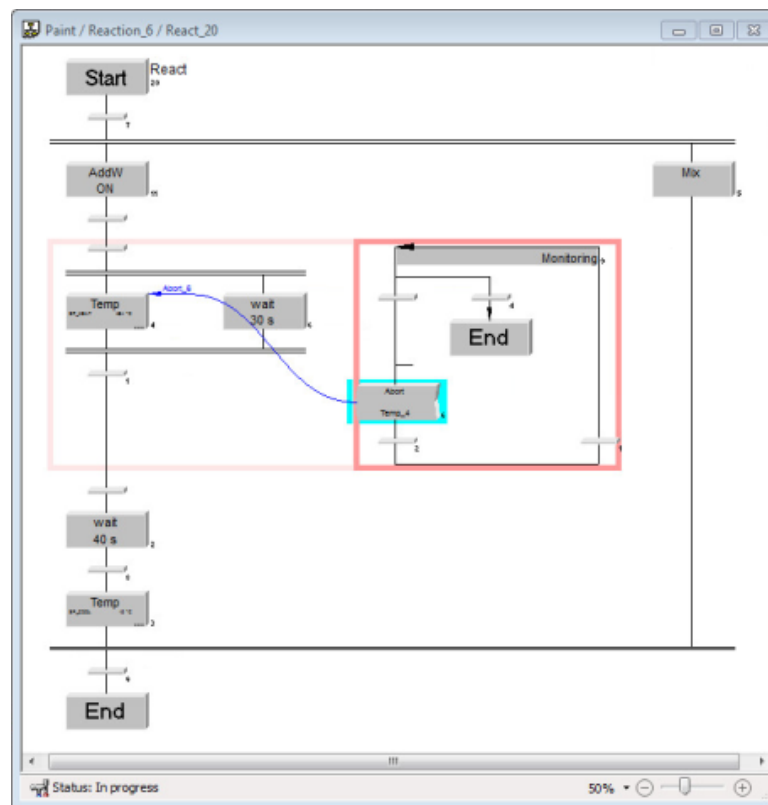


11.7 Command step

A command step is used to pass a command (ISA-88 command) in the recipe operation to one or more target recipe procedure elements (RPEs). This is very helpful in case of automatic exception handling if it can be determined upfront how the system should react to undesired process conditions.

A command step can also perform procedural actions across procedural element boundaries.

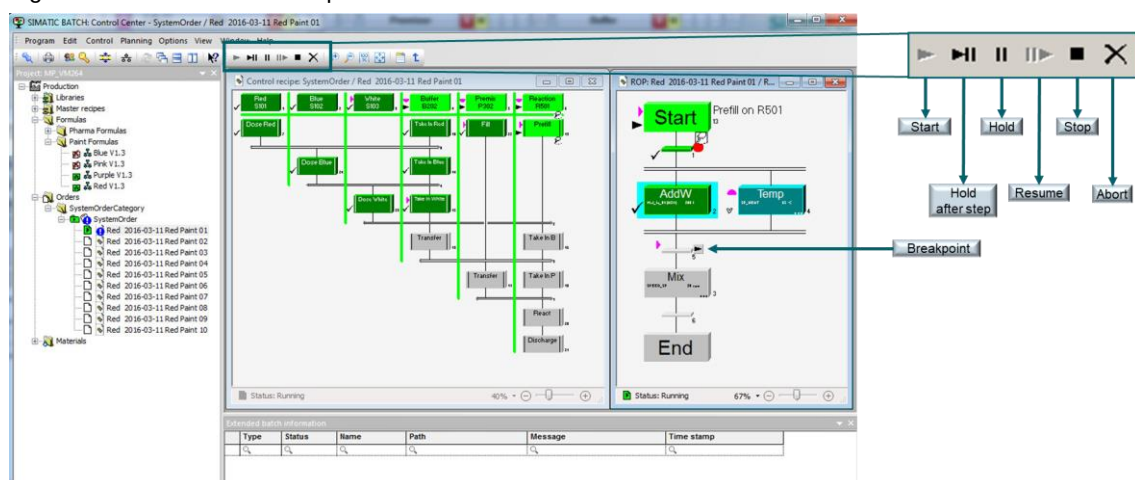
Figure 11-9 Example command step



11.8 Batch operation

At runtime of a batch the operator has many possibilities to monitor and control the batch, depending on the user rights. Several windows can be opened and arranged within the BATCH Control Center to monitor different sections of the control recipe. The operator can command the different recipe elements recipe procedure (RP), RUP, ROP, RPH and transitions. Recipe parameters can also be modified on any level and will be forwarded to the controller level.

Figure 11-10 Runtime batch operation



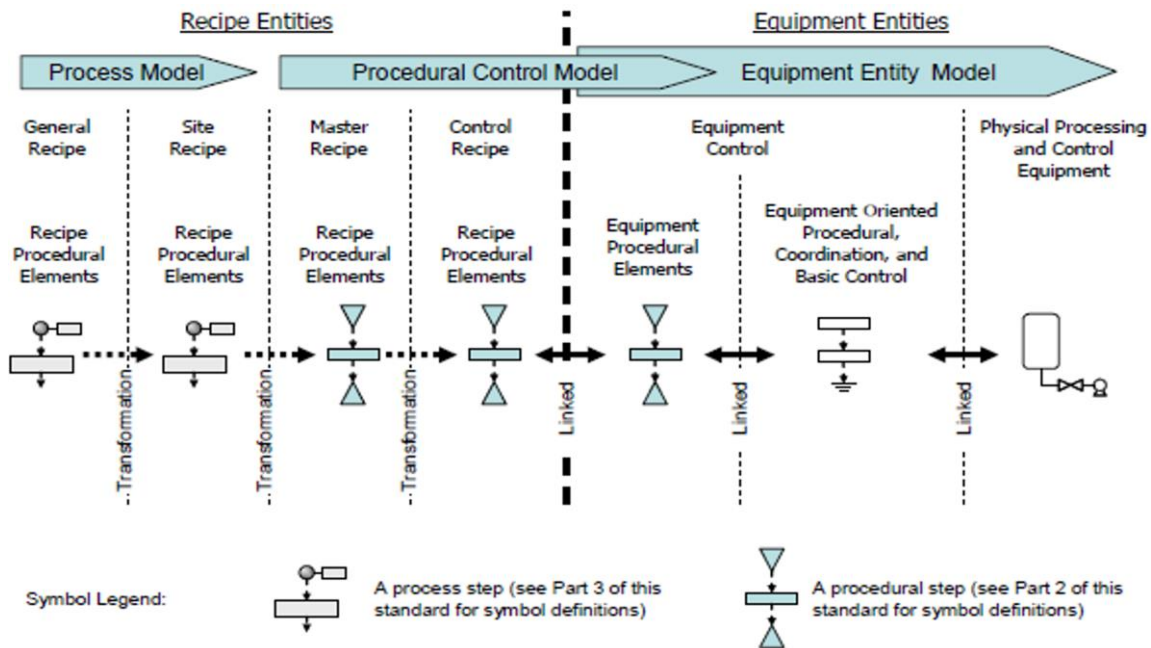
11.9 Batch Online Structural Changes

You can refine the master recipes in the test mode online structure change (OSC) and adjust them perfectly to the production processes. In the test mode OSC it is possible to modify recipe structures in planned, released and started batches. The altered or changed recipe structure in the control recipe (batch) can be saved for further use as a new master recipe at any time, serving a faster recipe development cycle.

12 Recipe transformation and transferability

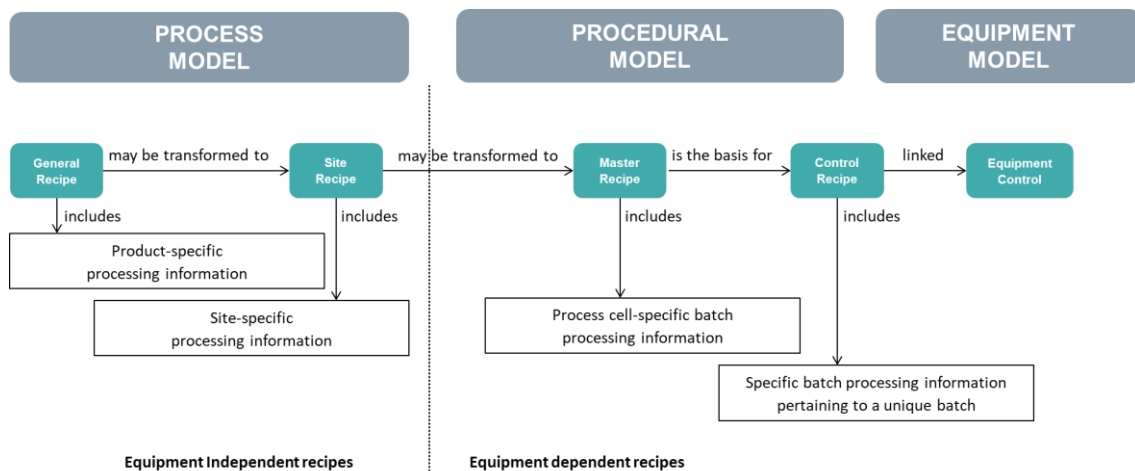
In the middle of the figure below from the ANSI/ISA-S88 part 1 (reference \4\)
“Information flow from general recipe to the equipment entity” the bi-directional linking between the recipe phases in the control recipes and the equipment phases in the controller is summarized. At the righthand side the equipment entity model is shown. These topics are described in detail in the previous chapters.

Figure 12-1 Information flow from general recipe to equipment entity



At the left-hand side, the figure shows the transformation steps for the recipe entities. Portability of the recipe was mentioned in section 8, where the generic equipment model approach was identified as the approach that could make the recipe less equipment dependent.

Figure 12-2 Relationship between four types of recipes in the ISA-88 standard



The ISA-88 standard defines four types of recipes, each having a different purpose from the others. The most important of these recipes are the general recipe and the master recipe.

General recipe

The general recipe, often referred to as an equipment-independent recipe, is generated for each product variant that is produced. The general recipe defines the material and process dependencies required to make the product in such a way that it is independent of the site at which the product is manufactured and the scale of manufacture, and specific equipment used. The general recipe is usually created during pilot plant scale up of a recipe used for R&D.

Site recipe

In addition to the general recipe, a site recipe may be maintained for each manufacturing site at which the product is made. The site recipe may specify local materials and may correspond to a general recipe that is modified for language and local units of measurement.

Master recipe

Then a level down, a master recipe is usually generated for each process cell and for each product variant, and may define exactly how the product is to be made in the process cell based on the units in the cell, material flows between units and the equipment available. Usually at this stage some equipment specifics come into the recipe.

Control recipe

Finally, a control recipe is generated for each batch and may describe custom options and formula values for a specific batch.

The conversion of general recipes to master recipes is documented in the part 3 of the ISA-88. Previously this was a manual operation, but this recipe transformation will be done more and more automatically via PLM tools. The advantage of using such tools is that recipe development and transformation must be standardized, which increases know-how and information transfer between the R&D and production departments.

Although it's possible to reuse master recipes across sites or process cells, their portability may be reduced because of equipment execution details. In such cases it is better to exchange the equipment independent recipes and transform these to master recipes with the unit specific equipment information. The generation of master recipes based on general/site recipes outside the master recipe editor is possible by defining the master recipes electronically based on the BatchML standard

To facilitate the import and export of recipe information to and from SIMATIC BATCH, a BatchML interface is available. BatchML provides a set of XML type and element definitions that are well suited for the exchange of ISA-88 based data. Siemens expanded the standardized BatchML specification to map all aspects of recipes in SIMATIC BATCH – in accordance with the BatchML standard. More details and some helpful examples can be found in this FAQ:

<https://support.industry.siemens.com/cs/us/en/view/109747147>

13 Interfaces towards other systems

SIMATIC PCS 7 and SIMATIC BATCH offer various interfaces for data exchange with batch execution related systems. It is important to use the correct interface for the intended use case.

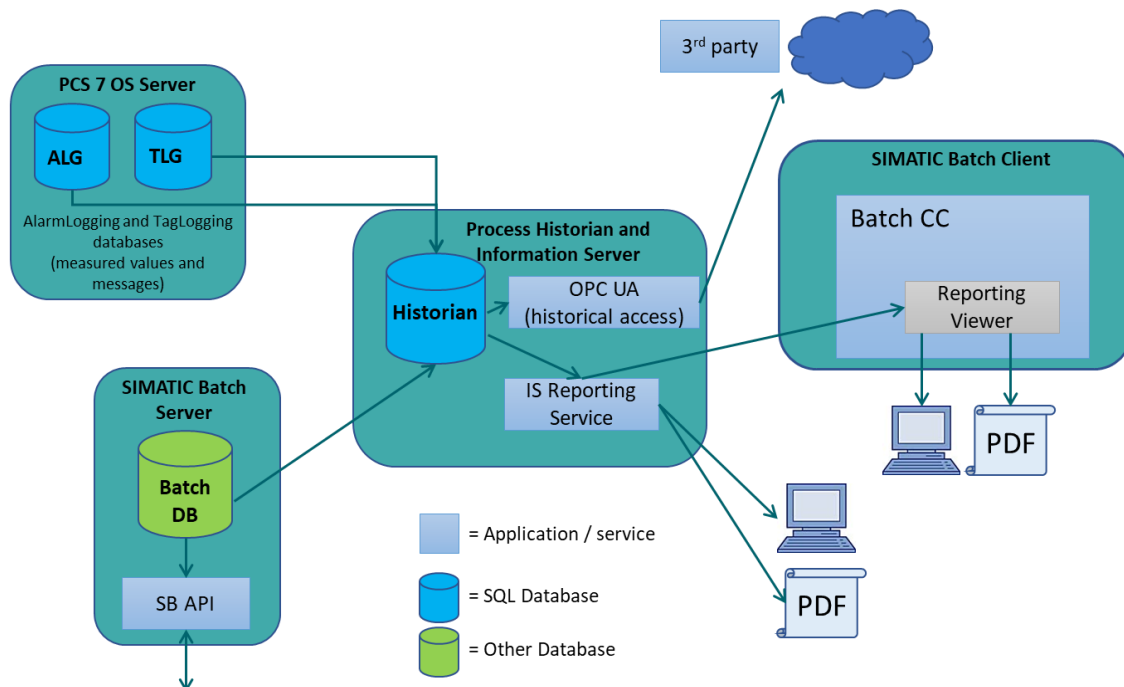
To support material transfers SIMATIC BATCH inherits the location model that is defined as part of the physical model in SIMATIC Manager. Locations are used as setpoints of recipe phases to define source, destination and via of a material transfer. If an equipment phase leverages SIMATIC Route Control for the material transfer then the location setpoints are automatically transferred to the route manager for allocation, control and monitoring of the control modules involved.

For reporting of batch-contextualized data, the reporting engine on the SIMATIC BATCH server can be used. Reports are based on Microsoft SQL Server Report Services (SSRS) and can easily be customized. See also

<https://support.industry.siemens.com/cs/us/en/view/109481496>

For monitoring, customized reporting and connection towards a 3rd party historian, it is recommended to use the SIMATIC Process Historian. The Process Historian (PH) collect runtime data from the PCS 7 OS and SIMATIC BATCH in near-real-time and stores this in a SQL database for long-time archiving. The process historian thus contains all process related data, contextualized as batch-relevant data and non-batch-contextualized data. The main dataflows of the process historian are shown here:

Figure 13-1 Dataflows of the process historian



Use of process historian for external access of data offers various advantages: the load on applications is reduced, redundancy issues are avoided, and the PH can be placed in the demilitarized zone (DMZ) for cybersecurity compliancy. The process historian offers various interfaces, not only SQL interfaces (commonly used for reporting and dashboarding) but also an OPC UA interface for connectivity to 3rd party historians and cloud-based data warehousing.

Note that the process historian only contains runtime data, such as control recipes. Master recipe data is maintained on the batch server and can be accessed using the batch API, which will be covered next.

SIMATIC BATCH offers a COM API interfaces that allows external applications to interact bidirectional with SIMATIC BATCH, for instance to launch a control recipe from a 3rd party MES engine, or to read details on master recipes. The API is described in this manual:

<https://support.industry.siemens.com/cs/us/en/view/92344632>

SIMATIC BATCH supports external recipe creation, by means of a BatchML interface, which was covered in chapter [12](#).

14 Acronyms

Term	Definition
AS	Automation System = PLC
CFC	Continuous Function Chart
CM	Control Module
CMT	Control Module Type
DCS	Decentralized Control System
DMZ	Demilitarized Zone
eBR	Electronic Batch Recording
EPH	Equipment Phase
EPHT	Equipment Phase Type
EM	Equipment Module
EMT	Equipment Module Type
EOP	Equipment Operation
IO	Input Output
MES	Manufacturing Execution System
OSC	Online Structure Change
P&ID	Piping & Instrumentation Diagram
PCell	Process Cell
PFD	Process Flow Diagram
PH	Process Historian
PLC	Programmable Logic Controller
RP	Recipe Procedure
RPE	Recipe Procedure Elements
RPH	Recipe Phase
ROP	Recipe Operation
RUP	Recipe Unit Procedure
SCADA	Supervisory Control And Data Acquisition
SFC	Sequential Function Chart Type

15 Appendix

15.1 Service and support

Industry Online Support

Do you have any questions or need assistance?

Siemens Industry Online Support offers round the clock access to our entire service and support know-how and portfolio.

The Industry Online Support is the central address for information about our products, solutions and services.

Product information, manuals, downloads, FAQs, application examples and videos – all information is accessible with just a few mouse clicks:

support.industry.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.

Please send queries to Technical Support via Web form:

support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page:

siemens.com/sitrain

Service offer

Our range of services includes the following:

- Plant data services
- Spare parts services
- Repair services
- On-site and maintenance services
- Retrofitting and modernization services
- Service programs and contracts

You can find detailed information on our range of services in the service catalog web page:

support.industry.siemens.com/cs/sc

Industry Online Support app

You will receive optimum support wherever you are with the "Siemens Industry Online Support" app. The app is available for iOS and Android:

support.industry.siemens.com/cs/ww/en/sc/2067

15.2 Industry Mall



The Siemens Industry Mall is the platform on which the entire Siemens Industry product portfolio is accessible. From the selection of products to the order and the delivery tracking, the Industry Mall enables the complete purchasing processing – directly and independently of time and location:

mall.industry.siemens.com

15.3 References, links and literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109784331
\3\	P. A. Sinclair and J. R. Savery, "Method of Generating Recipe for Process". U.S. Patent Patent 8,160,735 B2, April 17, 2012
\4\	International Society of Automation, "ANSI/ISA-88.00.01-2010 Batch Control Part 1: Models and Terminology," October 2020. [Online]. Available: https://www.isa.org/products/ansi-isa-88-00-01-2010-batch-control-part-1-models
\5\	International Society of Automation, "ISA95, Enterprise-Control System Integration," International Society of Automation, [Online]. Available: https://www.isa.org/standards-and-publications/isa-standards/isa-standards-committees/isa95 [Accessed October 2020]
\6\	Siemens, "SIMATIC PCS 7 Process Control System Compendium Part C - Equipment Phases with SFC Types (V9.0)," October 2018. [Online]. Available: https://support.industry.siemens.com/cs/us/en/view/109761602
\7\	Siemens, "SIMATIC Process control system PCS 7 SIMATIC BATCH V9.0 Getting Started," August 2020. [Online]. Available: https://support.industry.siemens.com/cs/ww/en/view/109781909 .
\8\	Siemens, "SIMATIC BATCH V9.0: BatchML reference & examples," April 2017. [Online]. Available: https://support.industry.siemens.com/cs/us/en/view/109747147
\9\	Siemens, "Sample application for SIMATIC BATCH COM API," December 2016. [Online]. Available: https://support.industry.siemens.com/cs/us/en/view/92344632
\10\	R. Peoples, "6 Steps to Designing a Flexible Control System with ISA-88," April 2014. [Online]. Available: https://www.crossco.com/blog/6-steps-designing-flexible-control-system-isa-88/

15.4 Change documentation

Table 15-1

Version	Date	Modifications
V1.0	01/2021	First version